UNIVERSITY OF LEEDS

KneeTrack Pro

Design File



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User Guide

User Guide

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1. Purpose

This device measures knee flexion angle in real time using a rotary encoder attached to the leg. It enables the examiner to mark specific flexion positions on the patella and record patellar displacement data across a range of motion. The output is a graph of patella medio lateral position vs knee flexion angle.

2. Equipment

- Arduino Nano 33 BLE with encoder controller firmware
- 128 PPR rotary encoder (attached to leg)
- 2x push buttons (Button 1 and Button 2)
- LED 1 (calibration indicator)
- LED 2 (recording indicator)
- Laptop with Arduino Serial Monitor (9600 baud)
- Skin marker pen

NOTE	Ensure the Serial Monitor is open on the laptop before beginning the procedure. Set baud rate to 9600.

3. Controls and Indicators

3.1 Buttons

Control	Action	Function
Button 1	Single press	Enter / exit Calibration Mode
Button 1	Double press (within 350ms)	Zero the encoder angle
Button 2	Single press	Start / stop Recording Mode

3.2 LED Indicators

LED 1	LED 2	Meaning
Off	Off	Idle / normal operation
On steady	Off	Encoder is within a calibration zone
Off	Off	In calibration mode, outside all zones
Flashes x2	Flashes x2	Calibration mode entered
Flashes x3	Flashes x3	Calibration mode exited
Rapid flash	Unchanged	Encoder successfully zeroed (2 seconds)
Off	On steady	Recording mode active

4. Measurement Procedure

Follow these steps in order for each patient measurement session.

Phase 1: Calibration and Zeroing

#	Action	Detail
1	**Connect to laptop**	Plug in the device via USB and open the Serial Monitor at 9600 baud.
2	**Position the leg**	Straighten the patient's leg fully to the anatomical reference position (0° extension).
3	**Zero the encoder**	With the leg fully straightened, double-press Button 1 (two presses within 350ms). LED 1 flashes rapidly for 2 seconds. Serial monitor prints: encoder count: [value], angle zeroed. The current position is now registered as 0°.

IMPORTANT	Zeroing must be performed with the leg fully straight. Any residual flexion at the time of zeroing will offset all subsequent angle readings.

Phase 2: Anatomical Marking

This phase uses the calibration zones to identify and mark specific flexion angles on the patella.

#	Action	Detail
1	**Enter Calibration Mode**	Press Button 1 once. LED 1 will now illuminate at 0°, 45°, and 90° (±tolerance).

2	**Begin slow flexion**	Slowly bend the knee from 0° toward full flexion. Maintain a steady, controlled pace.
3	**Mark at LED illumination**	Each time LED 1 lights up, palpate and mark a dot on the lateral patellar condyle using the skin marker. LED 1 indicates the knee is within a calibration zone (0°, 45°, or 90°). After drawing the dot, press Enter on your laptop and alter the RGB sliders until only the colour dot is visible – the dot should be circled and a 'Tracked' message should show.
4	**Apply colour key**	Use a different colour for each flexion angle according to your colour key. Record which colour corresponds to which angle.
5	**Exit Calibration Mode**	Once all three angles have been marked, press Button 1 once to exit. Both LEDs flash three times.

NOTE	Three dots should be marked in total: one at approximately 0°, one at 45°, and one at 90°. If LED 1 does not illuminate at the expected position, re-zero the encoder and repeat Phase 1.

Phase 3: Recording

#	Action	Detail
1	**Return leg to 0°**	Fully extend the leg back to the starting position.
2	**Confirm Serial Monitor open**	Ensure the laptop Serial Monitor is visible and ready to capture output.
3	**Start recording**	Press Button 2 once. LED 2 turns on and stays on. Serial monitor prints: Recording
4	**Begin controlled flexion**	Slowly and steadily flex the knee from 0° through to full flexion. The device will log a timestamped entry each time the encoder crosses 0°, 45°, or 90°.
5	**Monitor serial output**	Each crossing is printed in the format: [time_ms, angle]. For example: [1243, 0] / [2891, 45] / [4502, 90]. The medial lateral translation each point will also

		be exported to Excel in the format [time,ms,medial-lateral translation,mm]
6	**Stop recording**	Press Button 2 again once the range of motion is complete. LED 2 turns off. Serial monitor prints: recording finished

NOTE	Each angle crossing is recorded once per pass. If the encoder moves back and forth through a target angle, each pass in either direction will generate a new entry.

Phase 4: Data Analysis

#	Action	Detail
1	**Copy serial output**	Copy the timestamped data from the Serial Monitor to a spreadsheet or analysis tool.
2	**Add iPhone Recording data**	Copy the (time, translation) data recorded by the iPhone to the same analysis tool as the serial output.
3	**Align time and position data**	Correlate the timestamps from the serial output with the iPhone recorded dot positions to assign a mediolateral displacement value to each angle.
4	**Plot the graph**	Plot knee flexion angle (x-axis: 0°, 45°, 90°) against patella mediolateral displacement (y-axis, mm) to visualise patellar tracking across the range of motion.

5. Serial Monitor Output Reference

Output	When	Meaning
calibration mode	Button 1 single press	Calibration mode entered
calibration mode exited	Button 1 single press (again)	Calibration mode exited
encoder count: X, angle zeroed	Button 1 double press	Encoder zeroed at current position
Recording	Button 2 single press	Recording started, timer running
[time_ms, angle]	During recording	Encoder crossed 0, 45, or 90 degrees
recording finished	Button 2 single press (again)	Recording stopped

Example serial output for a complete recording session:

```
Recording
[843, 0]
[2104, 45]
[3891, 90]
recording finished
```

6. Troubleshooting

Problem	Likely cause	Action
LED 1 does not light up in calibration mode	Encoder not zeroed correctly	Re-extend the leg fully and double-press Button 1 to re-zero
LED 1 stays on continuously	Button wired incorrectly or stuck	Check button wiring; signal pin should connect to GND via button only

Angle readings drift over time	Encoder slipping on mounting	Re-secure the encoder and re-zero before the next recording
No output in Serial Monitor	Baud rate mismatch or wrong port	Set Serial Monitor to 9600 baud and confirm correct COM port
Double press not registering	Presses too slow	Press both clicks within 350ms, or adjust DBL_WIN in the firmware
Crossings missed during recording	Leg moved too quickly	Perform flexion more slowly to ensure the encoder passes through the ±2° tolerance window

7. Device Specification

Parameter	Value
Microcontroller	Arduino Nano 33 BLE (ATmega328P)
Power supply	3.3V via Micro USB
Encoder resolution	128 PPR
Angle resolution	2.8° per count
Calibration zones	-2 to 4° / 43 to 47° / 88 to 92°
Recording targets	0°, 45°, 90° (±1.5° tolerance)
Double press window	350 ms
Button debounce time	50 ms
Serial baud rate	9600

Instructions for Assembly and Disassembly

Instructions for Assembly

1. Assemble and adjust phone holder
    a. Insert the foot and claw into the phone holder
    b. Adjust width to the phone size of choice and screw claws in place [2x 3mm]
2. Fit hinge cap to brace
    a. Align hinge cap holes with the tapped holes on mental hinge
    b. Place and tighten 1x M3 screw
3. Assemble electronic housing
    a. Align the rotary encoder hole in the electronics closure [6mm], with the larger 6mm cavity in the hinge cap
    b. Insert rotary encoder through both holes and gently push the encoder to ensure it is seated securely
    c. Place Arduino nano on its designated supports and screw it in [4x 1.6mm]
    d. Place the enclosure cap onto housing and it should fit into place
    e. Screw in the cap [4x 1.6mm] to ensure electronics remain securely housed
4. Screw assembled phone holder into electronics housing
5. Insert your phone into the full assembled device

Instructions for Disassembly

1. Remove phone from assembled device
2. Unscrew phone holder from electronics housing with hex head screwdriver
3. Disassemble electronic housing
    a. Unscrew the enclosure cap [4x 1.6mm]
    b. Remove enclosure cap from housing
    c. Unscrew Arduino nano and set it aside [4x 1.6mm]
    d. Pop out LEDs and buttons using a screwdriver
    e. Gently move encoder, and flex brace from side to side until it comes loose and remove from housing
    f. Unscrew the hexhead screw from the base of the electronics casing with a heak head screwdriver.
4. Remove hinge cap from brace
    a. Unscrew device [1x M3]
5. Disassemble phone holder
    a. Unscrew claws [2x 3mm]
    b. Remove the foot and claw from the phone holder

Cleaning Protocols

Specified Cleaning Procedure

Purpose

This cleaning procedure outlines the steps required to clean the device for safe and effective use while maintaining performance, safety and hygiene standards. The device is skin contacting and designed to be cleaned with 96% ethanol.

Scope

This procedure applies to the cleaning of the device immediately after each patient. The device should always by sterilised between uses to ensure that no contaminants or debris remain on the surface

Responsibilities

- Operators – Responsible for following cleaning procedures to ensure the device is cleaned properly
- Quality Assurance – Responsible for reviewing and validating this cleaning procedure to ensure its adequacy
- Maintenance Personnel – Responsible for conducting cleaning validation tests

Cleaning Materials

1. 96% Ethanol – Commonly found in hospitals. The solution is effective in killing bacteria and viruses commonly found on skin-contacting devices
2. Clean, Lint-Free Wipes – Used to apply the ethanol and wipe down the device
3. Disposable Latex Gloves – To be worn during the cleaning process to prevent contamination

Cleaning Procedure

1. Preparation – Ensure device is disconnected from power source. Inspect the device for any visible contamination or debris
2. Glove Application – Wear disposable gloves to prevent direct contact with the device and any potential contaminants
3. Initial Wipe – Dampen clean, lint-free wipe with 96% ethanol. Gently wipe the entire surface of the device to remove any visible dirt or organic material. Use clean wipes and additional ethanol as required
4. Surface Cleaning – Ensure that all crevices, joints and accessible areas are thoroughly cleaned. Take particular care to access all skin-contacting faces of the device

5. Disinfection – Allow the device to remain damp with 96% ethanol for at least 2 minutes to ensure proper disinfection. Wipe down the device with a dry, clean wipe to remove excess ethanol
6. Drying – Allow the device to air dry in a clean, dry area, free from contaminants. Ensure that the device is completely dry before use or storage
7. Inspection – After cleaning, inspect the device for any signs of damage, wear or residue from the cleaning solution. If any issues are found, the device should be sent for further inspection or repair

Frequency of Cleaning

- After Each Use – Clean the device immediately after use to prevent any buildup of contaminants

Disposal of Cleaning Materials

Any wipes, gloves or additional materials used during the cleaning process should be disposed according to the specific institution's guidelines for medical waste disposal

Record Keeping

Cleaning logs should be maintained to record the date, time, operator and any issues encountered during cleaning. The cleaning records should be reviewed periodically for compliance and to assess the efficacy of the cleaning procedure

Safety Precautions

Always work in a well-ventilated area when handling ethanol, as it is flammable. Keep ethanol away from open flames or high temperatures

Material Specification and Sterilisation Evidence

The device will be manufactured from fused deposition modelled (FDM) polylactic acid (PLA). Table 1 details the key properties of FDM PLA, meeting the proposed requirements detailed in Table 1. In accordance with the ISO 10993, PLA must undergo cytotoxicity, sensitisation and irritation tests to ensure its applicability to the device (1).

Table 1 – Properties of FDM PLA at 100% infill density

Young's Modulus [MPa]	**Ultimate Tensile Strength [MPa]**	**Yield Stress [MPa]**	**Electrical Resistance [Ohms]**	**Cost [GBP/kg]**
3039 (2)	48 (2)	26 (2)	1.16 x 10^14 (3)	10 (4)

Under ISO 14937 and ISO 17664-1, a Class Im skin-contacting device does not require sterilisation but must be amenable to disinfection (5, 6). ISO 15883-1 requires specification of validated disinfection protocols (7). Table 2 demonstrates testing of medical grade disinfectant solutions. While 0.85% sodium hypochlorite achieved complete microbial reduction, it caused surface degradation. Ethanol achieved 98% disinfection with no observed material damage; therefore, it is specified as the recommended disinfectant. Infill density will be set to 100% to minimise porosity and prevent disinfectant from becoming trapped within the material.

Table 2 – Efficacy and Long-Term Effects of Common Hospital Disinfectants (8)

	Efficacy of Disinfectant (%)	**Effects of Prolonged Use (~24 hours)**
96% Ethanol	98	None Observed
70% Isopropanol	87	Melting of Structure
0.85% Sodium Hypochlorite	100	Surface and Filament Damage

Free Body Diagrams

Free Body Diagram of Mounting Device

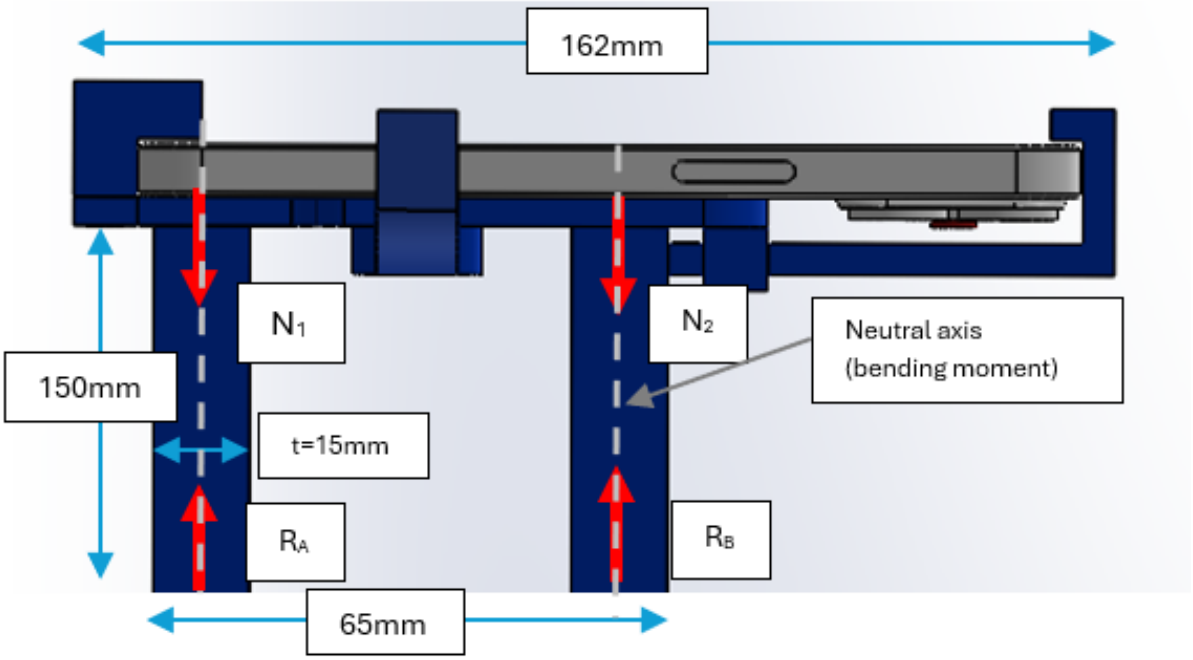


Figure 1 - Free body diagram showing forces acting on mounting device

Figure 1 shows a free body diagram of the forces acting on the mounting device. N1 and N2 are normal reaction forces, while Ra and Rb are supporting reaction forces acting at the base of the vertical supports. Assuming the worst-case scenario, the maximum stress in the system is 0.136MPa, significantly below the 26MPa yield stress of FDM PLA (2).

Assuming a worst-case scenario where the weight of the phone acts directly through the second vertical support:

$$Moment = mg * e\ (Eccentricity) = 0.204 \times 9.81 * 35.75 = 71.5\ Nmm$$

$$Second\ Moment\ of\ Area\ (Neutral\ Axis) = I = \frac{b * t^3}{12} = \frac{15 * 15^3}{12} = 4218.75\ mm^2$$

$$\sigma_{axial} = \frac{R_b}{A} = \frac{2.0}{225mm^2} = 0.0089\ MPa \qquad \sigma_{bending} = \frac{Mc}{I} = \frac{71.5 * 7.5}{4218.75} = 0.127\ MPa$$

$$\sigma_{max} = \sigma_{axial} + \sigma_{bending} = 0.136\ MPa \ll \sigma_{yield}$$

Failure Modes and Effects Analysis (FMEA) and Overall Safety

FMEA Table

Table 1 – Failure Modes and Effect Analysis for Device; showing Severity, Probability and Risk Priority Numbers before and after mitigation

Hazard	Effect	Causes	S	P	RPN	Reference	Recommended Mitigations	Validation	S	P	RPN
Software false output	Misdiagnosis	Inadequate or over filtering of data	3	4	12	ISO 62304 (9)	Apply frequency filtering; validate data against optical tracker	Ensure device accuracy ±3mm compared to optical tracker	3	2	6
Excessive strap pressure	Numbness, swelling, irritation	Overtighten brace straps by user	2	3	6	IEC 60601-1 (10)	IFU guidance on strap tightness	Comfortable Pressure: 10kPa	2	1	2
Bracket movement during motion	Invalid coordinate and position output	Device migration during knee flexion	5	3	15	ISO 14971 (11)	Reduce mass and centre of mass; minimise length of bracket	Repeatability testing across flexion cycles	5	2	10
Battery overheating	Potential burns or electrical fire	Short circuiting, PCB fault	5	3	15	IEC 62133I EC 60601 (11, 12)	Thermal testing; replace battery with USB cable	Electrical safety testing under simulated conditions	4	1	4
Adverse skin reactions	Dermatitis, irritation	Material sensitivity,	3	3	9	ISO 10993 (1)	Hypoallergenic; cleaning guidance	Cytotoxicity testing for prolonged exposure	3	1	3
Current leakage	Electrical shock risk	Insulation failure	4	3	12	IEC 60601-1; ISO 10993 (1, 10)	Redesign housing; use high resistivity material	Electrical safety testing under simulated conditions	4	1	4
Output misinterprets	Incorrect positioning data	Misaligned, calibration error	4	3	12	ISO 13485; ISO 14971 (13, 11)	Improved IFU, clinician training, useability testing	Formative useability validation passed	3	2	6
Mechanical failure	Patient injury, device breakage	Bracket fracture	3	2	6	ISO 14971 (11)	Fatigue testing, load-bearing specification	Mechanical endurance testing over 100 flexion cycles	3	1	3
Reuse contaminant	Infection transmission	Inadequate cleaning procedure	2	2	4	ISO 15883 (14)	Defined cleaning protocol	Cleaning efficacy testing	1	1	1

Child Safety Considerations

The device is intended exclusively for adult patients within a clinical environment, operated by a trained professional. Child access during normal use is therefore not anticipated. However, in accordance with ISO 14971, misuse scenarios must be considered across the product lifecycle. Given that the device contains small electronic components, wiring, and fastening screws, risk mitigation was conducted to address foreseeable scenarios through engineering and procedural means. Table 2 identifies the key hazards associated with child access, alongside the mitigations and relevant standards.

Table 2 – Risk mitigation table for child specific safety considerations

Hazard	**Risk**	**Mitigation**	**Standard**
Electronic components	Ingestion and electrocution	All components enclosed within PLA housing; Screwdriver required for disassembly	ISO 14971 (11)
Internal wiring	Electrocution	All wiring routed internally and kept out of reach of children	IEC 60601-1 (10)
M3 fastening screws	Ingestion/ choking	Screws packaged in sealed bags; Screw holes drilled precisely to ensure tight fit	ISO 14971 (11)
Liquid contact	Short circuiting	Instructions carry an explicit warning that the device must be kept away from all liquids and children	IEC 60601-1 (10)

By creating a fully enclosed casing and using high quality, non-conductive M3 screws, the design addressed the principal hazards associated with child access through engineering safeguards. Combined with warnings and recommended assembly precautions, the device was considered to meet the foreseeable misuse requirements outlined under ISO 14971.

Device Classification and Governing Regulations

The proposed design is a non-invasive, transient device intended for continuous use of less than 60 minutes. Under Annex IX of the Medical Devices Regulations Act of 2002, it is classified as a Class Im (measuring) device as it provides quantitative data to characterize patella movement, with intention to diagnose maltracking. In accordance with Annex I, a conformity assessment by a notified body is required to verify the accuracy of the system from sensor input to digital output. Annex II requires mitigation of risks that could compromise safety or performance. The device will also need to undergo UKCA certification to evidence that it meets requirements (15).

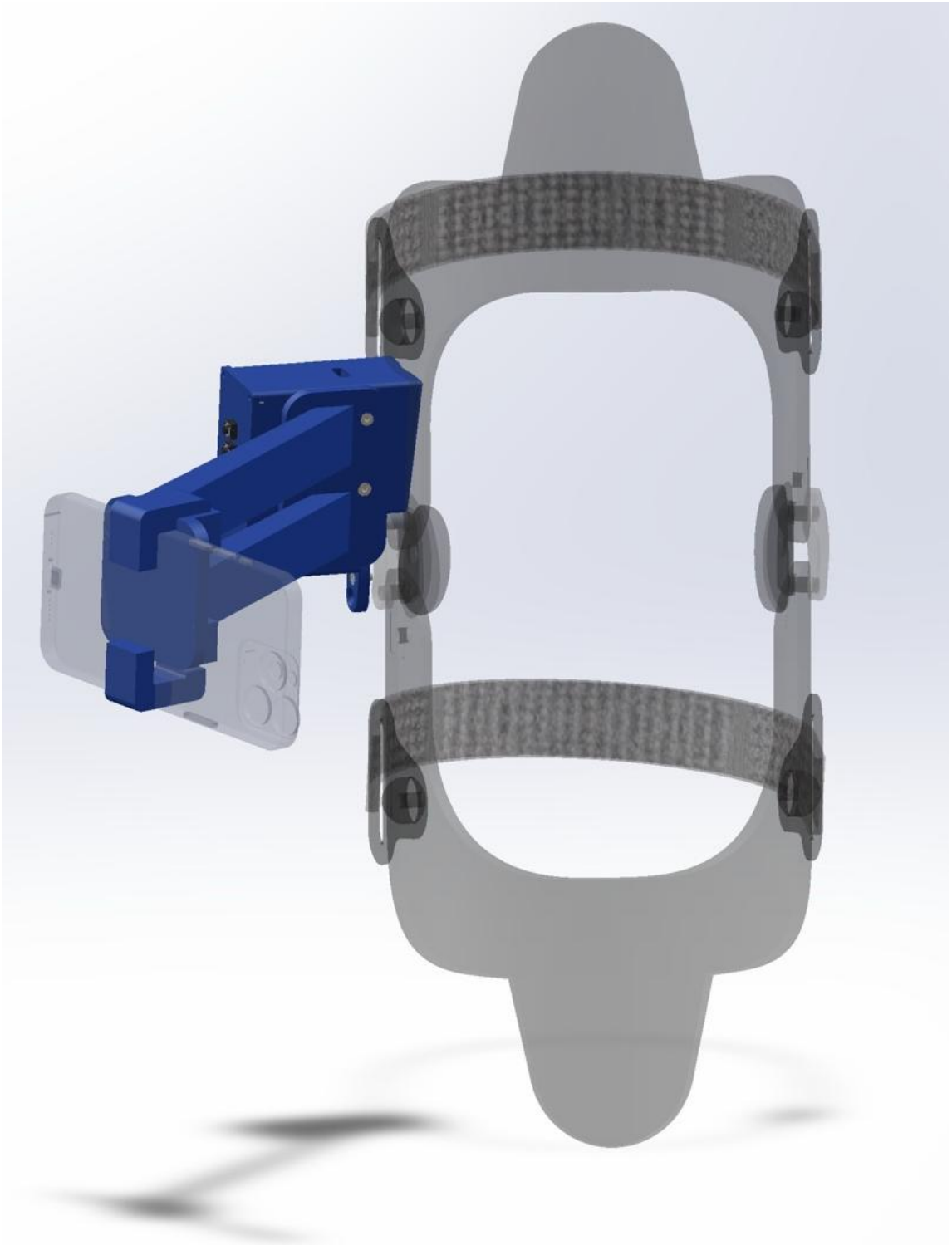
The manufacturer must demonstrate that the device performs its function while minimising risk. To ensure the device produces accurate diagnoses, the data collection and software must undergo verification and validation. To reduce error and improve the user experience, the design incorporates LED indicators to provide visual feedback. One LED indicates that the coloured markers have been identified, and the patient can begin rotating their leg. The second, linked to the rotary encoder, indicates the final angle at which the patient should stop rotation. This prevents measurements being taken from an uncalibrated reference, leading to incorrect completion of the diagnostic function.

Computer Aided Design (CAD) Illustrations

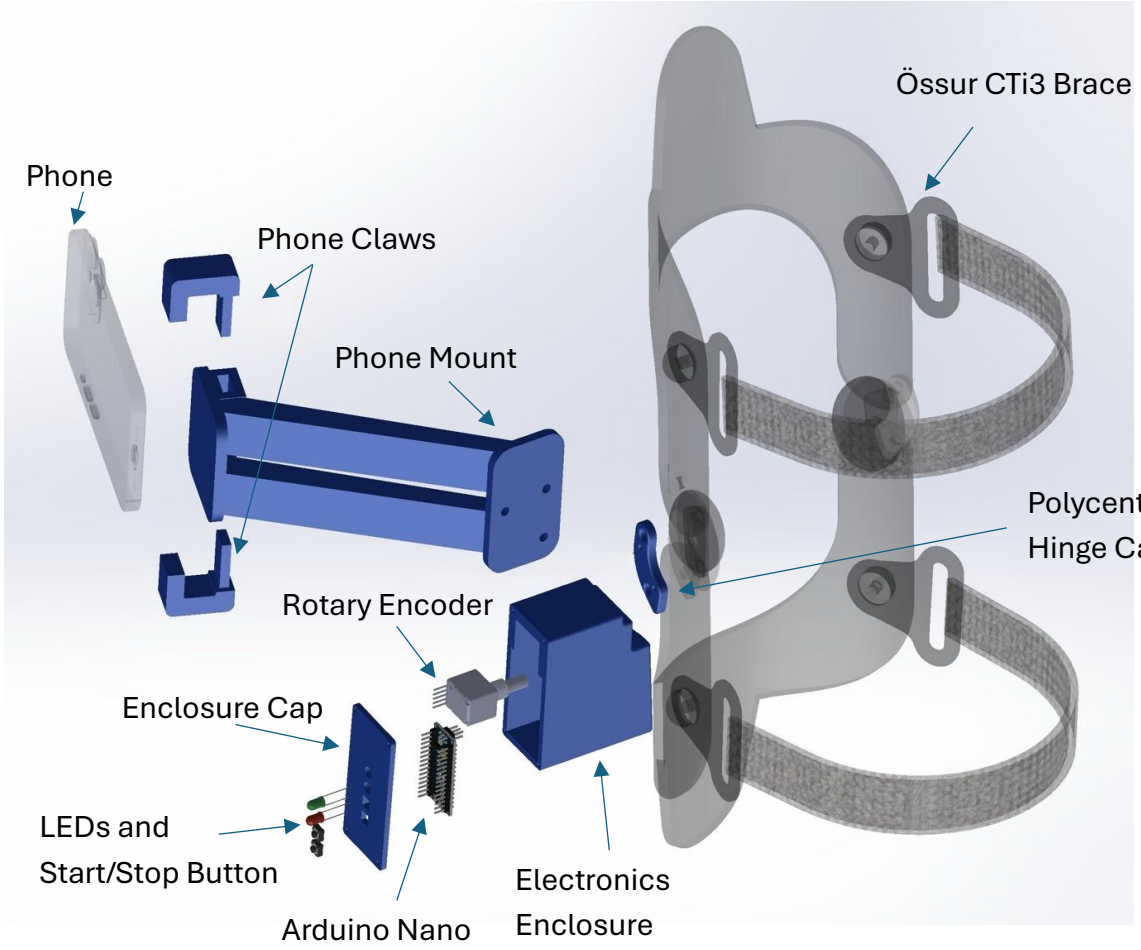
View in Use

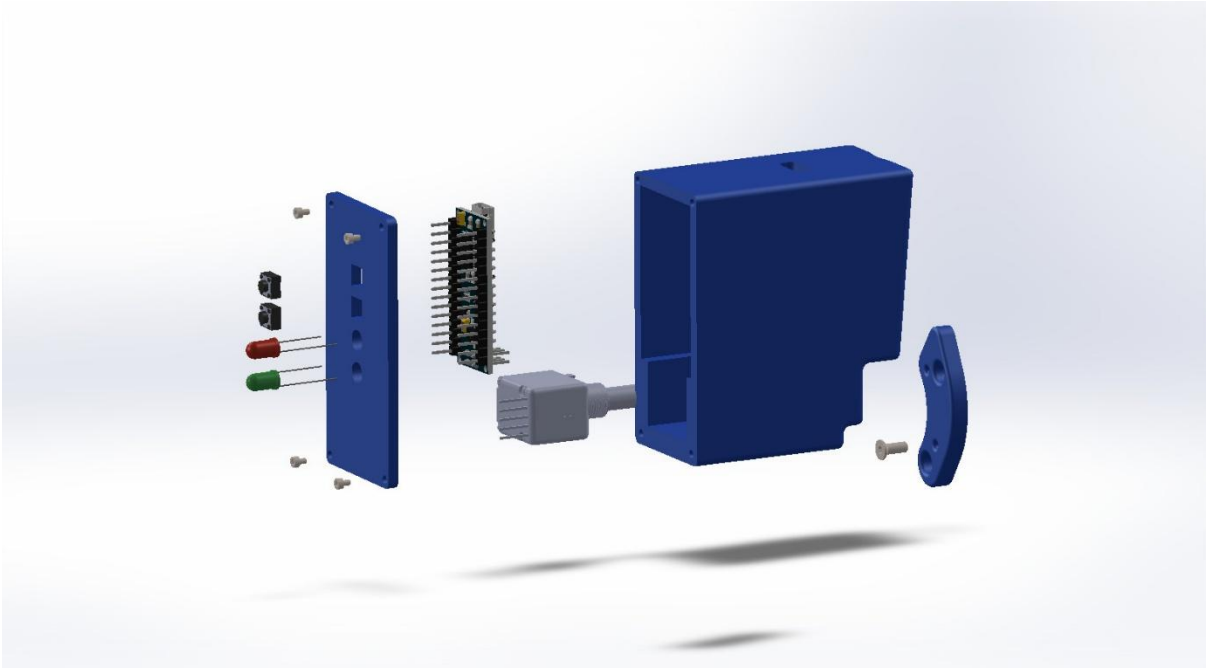


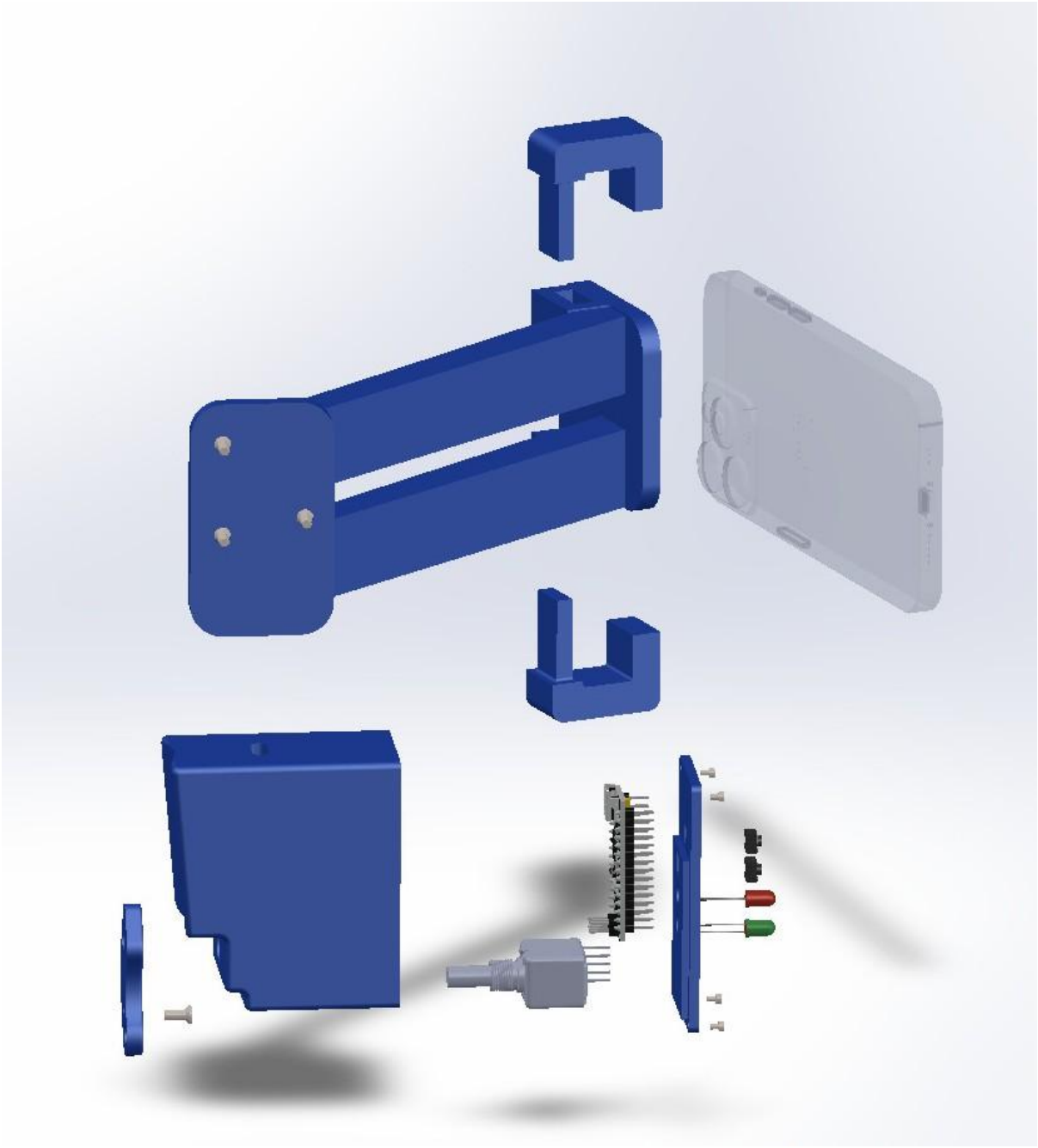
Isolated Views of Device



Exploded View







Brace Hinge Dimensions

There were no available dimensions for the polycentric hinge cap for the Össur CTi3 brace, to mitigate this, the hinge cap was traced onto a sheet of paper and measurements were taken from this instead.



Final Product

The final product mounts an electronics enclosure and phone stand to an Ossur CTI knee brace





Testing

Battery Testing

Introduction

In order to mitigate the trip hazard associated with using a USB cable to power the Arduino, testing was undertaken to determine if a 3.7 V, 1450 mAh lithium-ion battery (shown in Figure 1) would be safe for use in the system. The primary consideration was surface temperature, as a temperature of 40ºC contacting the skin is generally perceived to be warm and reaching the point of discomfort (16).



Figure 2 - 3.7 V 1450 mAh Lithium-Ion Battery

Methods

The first test undertaken was a simple voltage probe, to ensure that the voltage in the battery matched the specified value of 3.7 V. Voltage was measured three times, using the direct voltage setting of a multimeter, the mean of these values was 3.95 V. This is within a reasonable range, as the battery specification stated that it could be charged to a maximum value of 4.2 V.

The second test considered the surface temperature of the battery during charging. The battery was connected to a power supply, with current limited to 300 mA and voltage to 4.2 V, within the safety specifications for the product. The surface temperature was recorded every 30 seconds using a temperature probe.

Results

The results of the surface temperature test are shown in Figure 2 and Table 1.

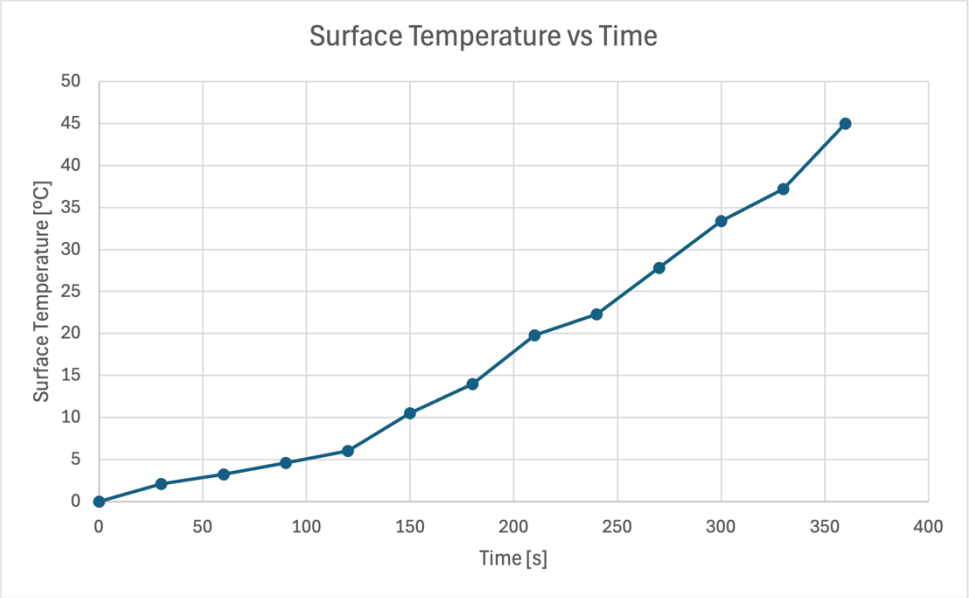


Figure 3 - Graph illustrating battery surface temperature against time

Table 1 – Raw data taken during experiment to consider battery surface temperature against time

Time (s)	Surface Temperature (ºC)
0	0.0
30	2.1
60	3.2
90	4.6
120	6.0
150	10.5
180	14.0
210	19.8
240	22.3
270	27.8
300	33.4
330	37.2
360	45.0

After 6 minutes, the surface temperature of the battery had reached 45ºC, exceeding the threshold for comfortable skin contact. This would be unsafe during operation, and the test was terminated at this point for safety reasons.

Conclusion

It was decided that a USB cable should be used to power the Arduino, as the risk of tripping over the cable had less severe implications than the potential for skin burns from an overheating battery. This aligns with international safety standards and ensures patient comfort. No further battery testing was carried out beyond this point, as the method was deemed unfit for purpose.

PLA Four-Point Bending Testing

Introduction

The phone bracket in the design is composed entirely of Fuse Deposition Modelled (FDM) Polylactic Acid (PLA). The bracket allows an iPhone 14 Pro to be mounted at a fixed distance of 15 cm from the patient's patella. A four-point bending test was conducted on a 10x10x100 mm specimen of 100% infill density FDM PLA to experimentally determine the flexural yield stress. This allows the structural integrity of the design to be considered, ensures device accuracy and aligns with guidelines established in ISO 10328 (17).

Methods

A four-point bending test was setup on an Instron 3000. The initial positioning is shown in Figure 1, a 32:64 mm ratio was used, and the sample was positioned such that the load application would be symmetrical through the material. Load was progressively applied from 0 to 2 kN, with the final shape of the sample shown in Figure 2.

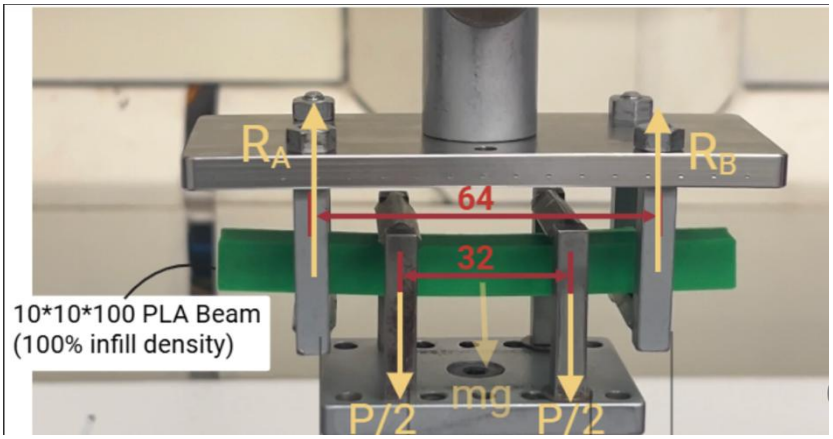


Figure 4 - Setup of PLA sample in Instron 3000

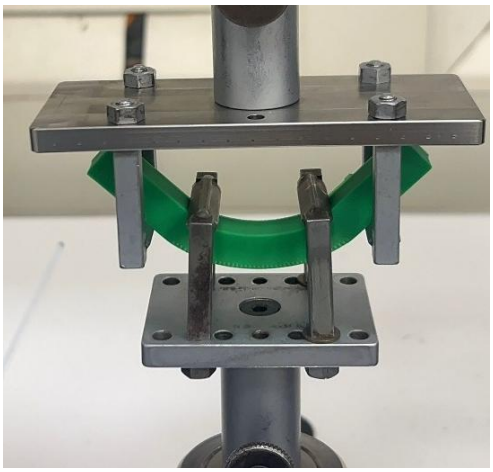


Figure 2 - Final shape of PLA sample in Instron 3000 after 2 kN loading

Results

Figure 3 details the relationship between force and displacement for the sample:

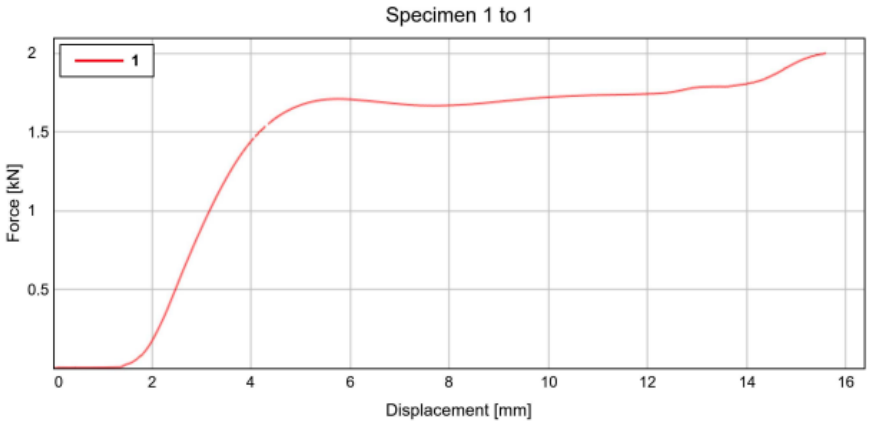


Figure 3 – Graph showing force against displacement for the four-point bending test

Based on theoretical equations (1.1, 1.2) and considering the peak of 1.706 kN at 5.52 mm, the flexural yield stress and engineering strain at this point can be calculated. The yield stress was found to be 81.9 MPa. Additionally, Figure 4 was produced, highlighting the relationship between stress and strain throughout testing.

$$\sigma = \frac{Mc}{I} = \frac{\frac{1706}{2} \times 16 \times 5}{833.3} = 81.9\ MPa$$

$$\varepsilon = \frac{6d\delta}{L^2} = \frac{6 \times 10 \times 5.52}{64^2} = 0.081$$

Equations 1.1, 1.2 – Engineering flexural stress and strain for four-point bending

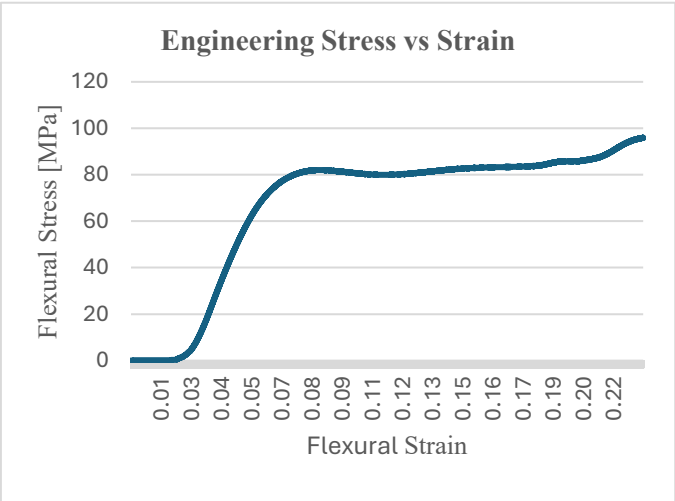


Figure 4 - Graph showing flexural stress against strain

The maximum stress in the sample was calculated based on a free body diagram of the bracket, shown in Figure 5, assuming a worst-case scenario in which the weight of the phone acts directly through the second vertical support of the bracket.

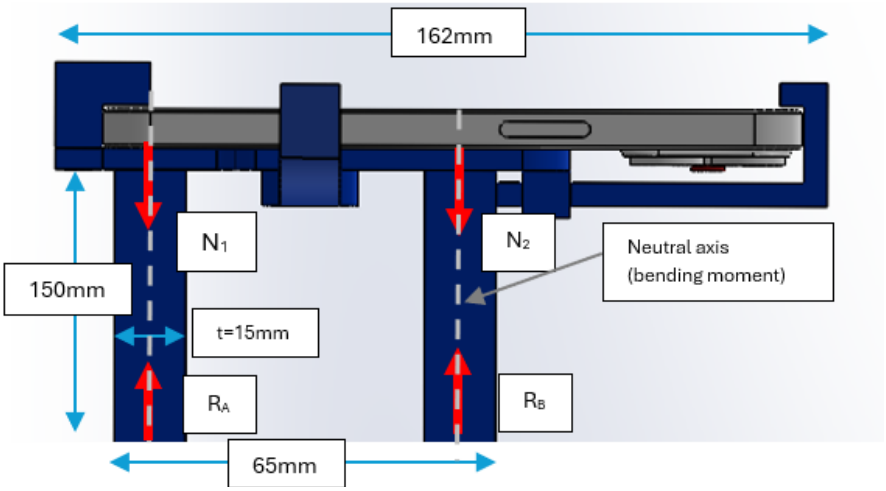


Figure 5 - Free body diagram showing forces acting on mounting bracket in worst case scenario

$$Moment = mg * e\ (Eccentricity) = 0.204 \times 9.81 * 35.75 = 71.5\ Nmm$$

$$Second\ Moment\ of\ Area\ (Neutral\ Axis) = I = \frac{b * t^3}{12} = \frac{15 * 15^3}{12} = 4218.75\ mm^2$$

$$\sigma_{axial} = \frac{R_b}{A} = \frac{2.0}{225mm^2} = 0.0089\ MPa \qquad \sigma_{bending} = \frac{Mc}{I} = \frac{71.5 * 7.5}{4218.75} = 0.127\ MPa$$

$$\sigma_{max} = \sigma_{axial} + \sigma_{bending} = 0.136\ MPa$$

This suggests that the bracket is at minimal risk of structural failure of fracture, as the maximum operating stress of 0.136 MPa is significantly below the experimentally determined flexural yield stress of 81.9 MPa for 100% infill density FDM PLA. This provides a substantial safety factor of 10, confirming the safety of the device.

Conclusions

The four-point bending test confirmed that the phone mounting bracket could safely be manufactured using FDM PLA. The flexural yield stress of 81.9 MPa is significantly larger than the predicted maximum operating stress of 0.136 MPa, providing safety factor of 10 and suggesting that the device is unlikely to undergo structural failure during operation. This satisfies the criteria laid out in ISO 10328, requiring the yield strength of the material to be at least twice the operating stress. Therefore, the bracket can be considered safe for use.

Field of View (FOV) Testing

Introduction

The patella tracking device design relies on the rear camera of the phonr capturing the marked colour points on the patient's knee throughout the full range of knee flexion. As the knee flexes, the marked points move within the captured image. Therefore, the camera's field of view (FoV) at the working distance must be large enough to always keep the points in the frame. This test measures how the width and height of the camera's FoV changed with distance from the subject. The results of the test informed the design of the brace mount that holds the phone. The distance from the phone to the knee was chosen to ensure full range of motion is captured when the brace is worn.

Method

The phone was mounted facing a flat, plain wall, with the camera orientated parallel to the wall surface. The phone was placed at a series of known distances from the wall, measured using a ruler, in 5cm increments from 15cm to 45cm, the setup is shown in Figure 1.

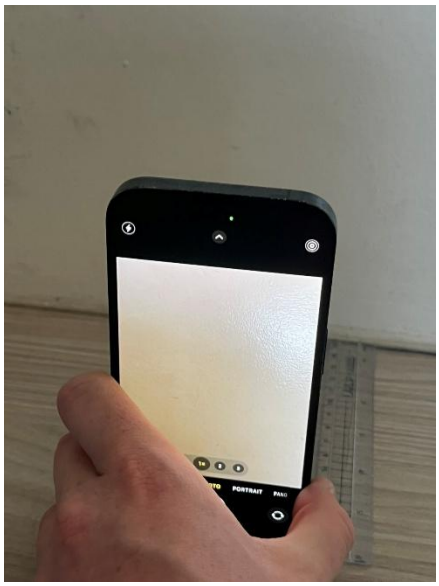


Figure 1.1 – FoV Accuracy Test Setup

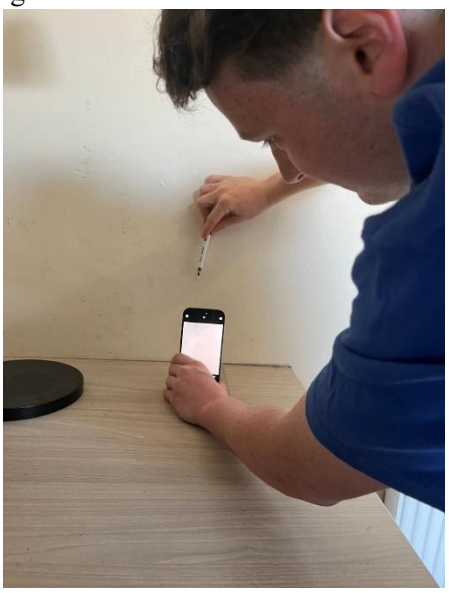


Figure 1.2 – Team member performing test

At each distance, the live camera feed was viewed on the connected laptop, and a point was marked on the wall at each of the four corners of the FoV. The width and height of the rectangle formed by these four marks were then measured using a ruler. This was repeated at every distance, and results were plotted to show how the width and height of the frame vary with distance from the subject.

Results

The results of the FoV test are shown in Figure 2 and Table 1. Both the frame width and the frame height increase approximately linearly with distance from the subject, as expected for a camera with a fixed angle of view. The frame height is consistently larger than the frame width, which is expected

since the camera was operated in portrait orientation. At 15cm the FoV is approximately 9.1cm x 14.25cm, this grows to approximately 25.4cm x 42.5cm at 45cm from the wall.

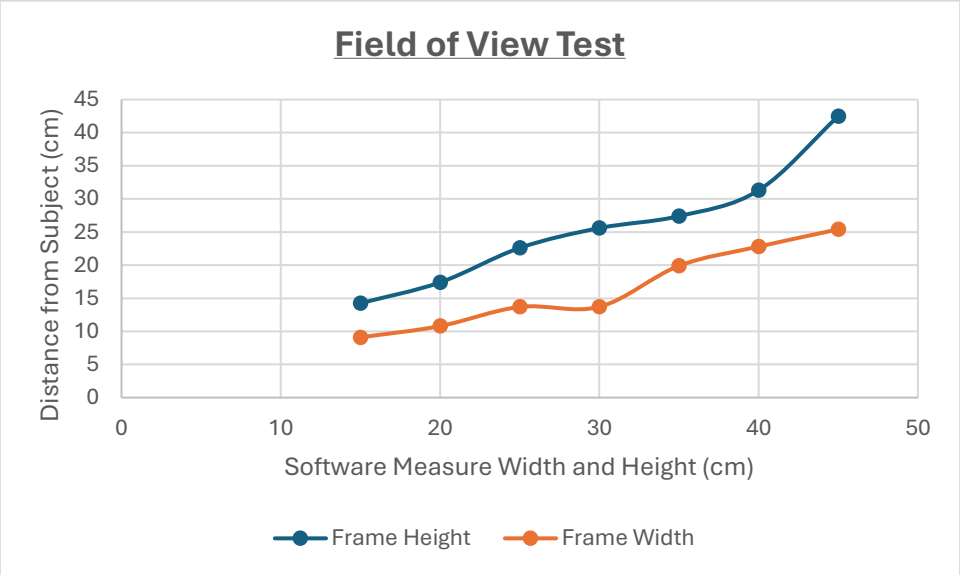


Figure 2 – Field of View Test Results

Table 1 – Raw data from the field of view test

Distance from Subject (cm)	Frame Width (cm)	Frame Height (cm)
15	9.10	14.3
20	10.8	17.4
25	13.7	22.6
30	13.7	25.6
35	19.9	27.4
40	22.8	31.3
45	25.4	42.5

The chosen distance value should be as small as possible, to reduce the magnitude of the bending moment acting on the mount. The frame width required was 1.93cm, the mean mediolateral range of patella displacement seen in the literature (18). As a result, 15cm was selected as the ideal distance for mounting the phone.

This distance allows the marked points on the knee to remain within the frame throughout the full range of motion. Sources of uncertainty in this test include the precision of the ruler used to measure the FoV corner, it is not certain the points are drawn at perfect right angles from each other. There is also difficulty in identifying the exact pixel at which the FoV ends, the corner marks were judged by eye from the live camera feed, this is likely the largest source of error. The measured values are therefore approximate, but the change in FoV for the tested range provides a good approximation when the mounting distance of the phone was determined.

Conclusion

The camera's field of view scales approximately linearly with distance from the subject, growing approximately 9.1cm x 14.25cm at 15cm to 25.4cm x 42.5cm at 45cm. These values gave a clear basis for choosing the mounting distance of 15 cm for the phone in the tracking device, which ensures the marked points of the patient's knee remain in frame throughout flexion. The frame width at this distance also covers the required 1.93cm of mediolateral patella displacement recorded in the literature.

Depth Sensor Accuracy Testing

Introduction

The patella tracking device relies on accurately identifying the marked colour points on the patient's knee. As the leg flexes, the patella (and the marked points) will move towards and away from the camera. It is important the phone's depth sensor can accurately determine the distance of the points from the sensor as the subsequent mediolateral values are reliant on this information. The iPhone 14 Pro was used in this test, which uses a Light Detection and Ranging (LiDAR) scanner to determine depth, so this test effectively assesses the reliability of this sensor.

Method

The phone was mounted facing a flat, plain wall, with the rear camera orientated parallel to the wall surface. The phone was placed at a series of known distances from the wall, measured using a ruler, in 5 cm increments from 15 cm to 45 cm, the setup can be seen in Figure 1.

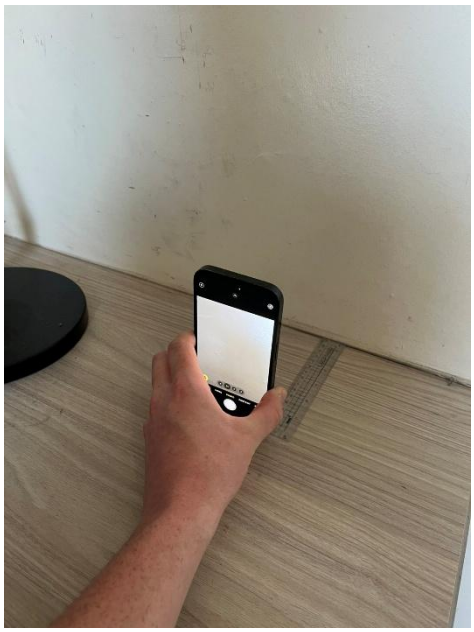


Figure 1.1 – Depth Accuracy Test Setup

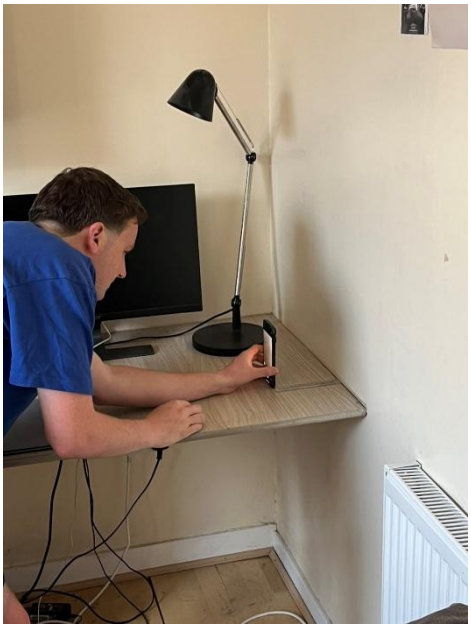


Figure 1.2 – Team member performing test

At each distance, the depth information from the sensor was captured using a Python script running on a connected computer, which saved the depth information of each pixel in the frame in an Excel spreadsheet. Since the wall was the only object in the camera's field of view, every pixel represented the same target distance. For each distance, the mean pixel depth of all the pixels was taken as the sensor-measured distance. The sensor measured value was then plotted against the ruler measured 'actual' distance to assess the sensor's accuracy.

Results

The results of the depth accuracy test are shown in Figure 2 and Table 1. Across the full range tested, the sensor measured distance was almost identical to the actual distance, with a maximum deviation of 0.3cm and a mean error of 1.013mm. The results show the sensor is accurate at all the measured distances and does not exhibit scaling error within this interval.

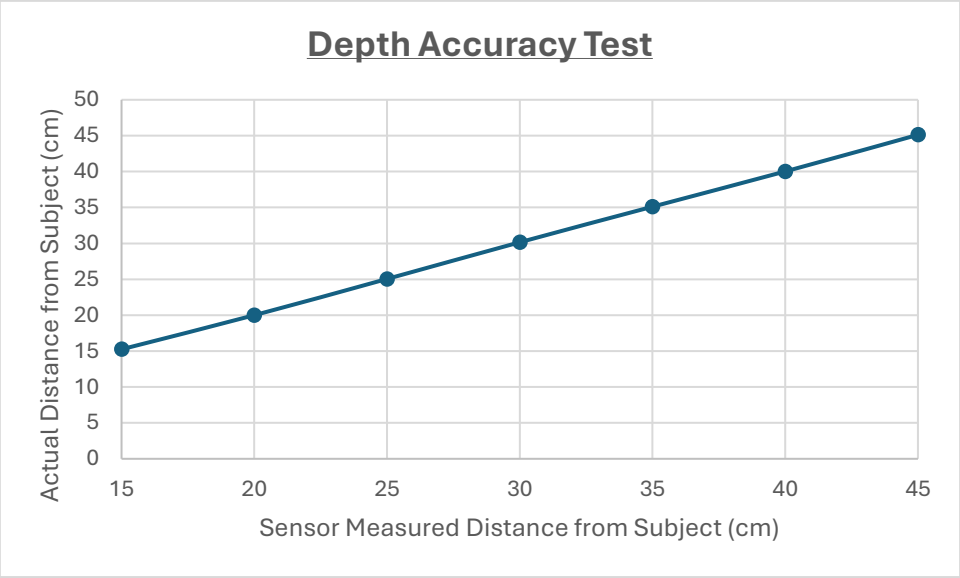


Figure 2 – Depth Accuracy Test Results

Table 1 – Raw data from the depth accuracy test

Actual Distance (cm)	Sensor Measured Distance (cm)	Absolute Error (cm)
15	15.25	0.25
20	20.00	0.00
25	25.05	0.05
30	30.16	0.16
35	35.11	0.11
40	40.02	0.02
45	45.12	0.12

The observed errors are an order of magnitude smaller than those reported in published studies of the same sensor at greater distances. For distances of over 10m, reported errors increase to around 16cm (19). The literature also showed that accuracy was best at short ranges with the device held still, identical to the conditions used in this test and the conditions that will occur during use of the KneeTrack Pro. Minor sources of uncertainty in the conducted test include slight tilt of the phone relative to the wall and the precision of the ‘actual’ ruler measurement. However, given the largest deviation observed was 0.3 cm, these effects can be thought to have had a negligible effect on the conclusion.

Conclusion

The iPhone's depth sensor measures distance with an error of ±1.013mm over the 15-45 cm range relevant to the patella tracking device. This is well within the accuracy needed the track the position of the marker of the patella during knee flexion. No further calibration of the iPhone sensor was deemed necessary before implementing the full tracking code.

Skin Marking Pen

To comply with ISO 10993 (1), any pen used to mark the patient's skin in a clinical setting is required to meet biological evaluation criteria for skin contact use. Standard retail pens and markers do not satisfy these requirements. Therefore, medically certified skin marking pens were selected for use in combination with this design. They are commonly used for ear piercing, Botox procedures and marking during surgery.

The Viscot EX Removable Inl DER Marker was identified as a suitable pen, recommended for use with the KneeTrack Pro based on suitability for medical applications and availability in multiple ink colours. Multiple colours are a key requirement, as the code relies on colour masking to isolate the specific anatomical points on the knee. According to the manufacturing data sheet, the ink contains flammable components and is required to be stored away from heat sources, used in well-ventilated areas and kept away from eyes. This is appropriate for clinical settings, which will likely already have eyewash stations and ventilated rooms suitable for handling ethanol.

Rotary Encoder

Device: ENA1J-B28-L00128L | **Resolution:** 128 PPR |

1. Introduction

The ENA1J-B28-L00128L incremental rotary encoder specifies a resolution of 128 pulses per revolution (PPR), corresponding to an angular resolution of 2.8125° per pulse. The purpose of this test was to verify the accuracy of the encoder output against independently measured angular positions, thereby confirming whether the manufacturer datasheet specification is reliable for use in precision measurement or control applications. Testing was conducted by mounting a 3D-printed pointer tip onto the encoder shaft and manually rotating it to target angles measured using a protractor. The encoder pulse count was read via software and converted to degrees using the relationship: Angle = (Pulses / 128) × 360°.

2. Results

Table 1: Comparison of measured angle and encoder-derived angle across the full rotation range.

Trial	Measured Angle (°)	Encoder Pulses	Encoder Angle (°)	Error (°)
1	30.0	11	30.94	+0.94
2	60.0	21	59.06	-0.94
3	90.0	32	90.00	0.00
4	120.0	43	120.94	+0.94
5	150.0	53	148.59	-1.41
6	180.0	64	180.00	0.00
7	210.0	75	210.94	+0.94
8	270.0	96	270.00	0.00
9	330.0	117	328.59	-1.41
10	360.0	128	360.00	0.00

Mean absolute error: 0.62° | **Maximum error:** 1.41°

3. Conclusion

The results demonstrate that the ENA1J-B28-L00128L encoder performs in close agreement with the manufacturer datasheet specification of 128 PPR. The observed errors are consistent with quantisation effects inherent to the 2.8125° per pulse resolution, rather than any device fault; angles that are exact

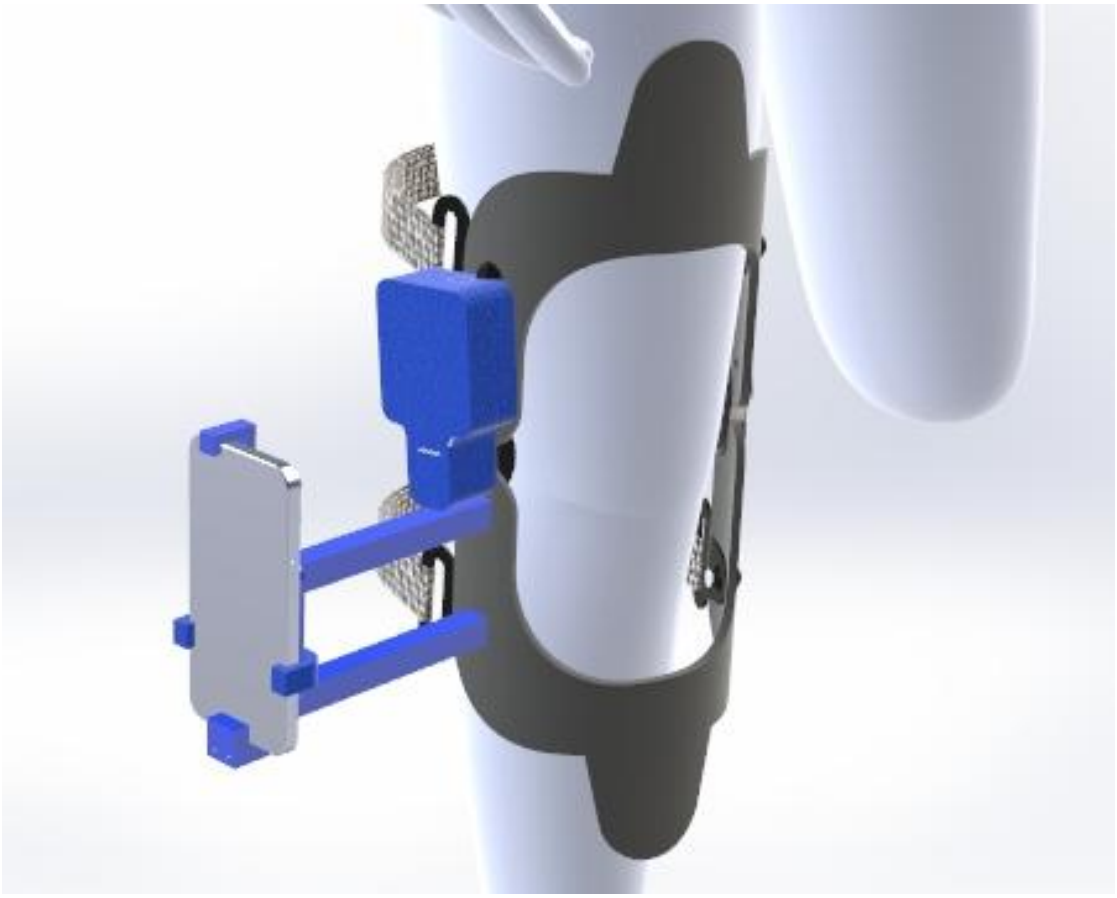
multiples of 2.8125° yielded zero error, while non-multiples produced rounding discrepancies bounded within one pulse width (±2.8125°). The maximum recorded error of 1.41° falls well within this theoretical limit. It is concluded that the encoder output is valid and the datasheet specification is accurate for the tested range of motion.

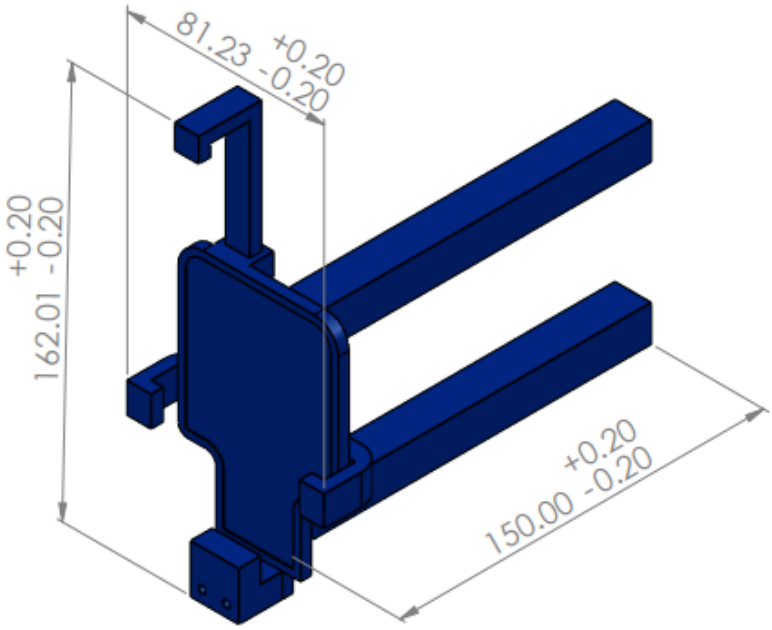
Evidence of Cyclic Design Process

Original Design Specification

User	Need	Implication	W	References	Verification	Design Requirement
Clinician	Quantify patella motion	Patella position measurable at a given time	5	Systematised MRI evaluation method (20)	Compare measured trajectory to MRI and motion capture data	Patella tracking accuracy ±3mm over full range of motion
Clinician	Full flexion motion capture	Knee within FOV and minimum depth	5	ISO 20685-1; Intel LiDAR Data Sheet (21, 22)	Minimum Separation = 15cm; Trigonometry at FOV	Sensor distance > 15cm
Clinician	Record flexion angle	Real-time joint angular measurement	4	Encoder Datasheet (23)	Calibrate potentiometer at 0º, 45º and 90º	Calibrated potentiometer and microcontroller
Clinician	Clear data display	Interpretable real-time output	4	ISO 9241-210 (24)	Practitioner feedback; Iterative design	Display of angle and patella position (<0.5s latency)
Clinician	Hygienic and reusable	Disinfectant resistant; Cleanable	4	ISO 17664-1; Datasheets (25,26)	Disinfectant exposure testing	Detachable, components can be disinfected
Clinician	Hinge aligned to TEA	Alignment identifiable and testable	4	Anatomical Literature (27)	Visual marking; Alignment jig test	Hinge alignment within ±2mm of TEA
Clinician	Measure across flexion	Hinge moves at least 90º	3	-	Angular limit testing	LED indicates rotation beyond 0-90º
Patient	Universal, comfortable fit	Accommodate 5th – 95th percentile	4	BS EN ISO 7250 (28)	Thigh and calf circumference measurement	Adjustable range of ±30mm for both circumferences
Patient	Prevent translation	Maintain fixation	4	-	Dynamic fitting test over full range of motion	Velcro fixation straps
Patient	Non-invasive	No irritation or skin damage	4	BS EN ISO 10993-10; Datasheets (1, 26)	Abrasion tests; Soft tissue phantom; Swelling check	Hypoallergenic material
Patient	Structural integrity	Withstand peak stresses	3	BS EN ISO 10328 (29)	Four point bending test; Physical stress test	Yield strength > 2 x Expected Operating Stress
Patient	Free movement	Consider biomechanics	3	ISO 6385 (30)	Gait comparison with and without brace	Mass < 0.5kg
Hospital	Traceability	Product ID	5	ISO 13485 (13)	ID readability check	Unique ID number
Hospital	Minimal training	Clear usage guidance	2	IEC 62366 (31)	Practitioner feedback; Iterative design	Concise step-by-step guide
Hospital	Easy end-of-life	Separable components	2	ISO 15223-1 (32)	Timed dismantling	Modular, separable design

Initial Design Ideas




81.23 +0.20 -0.20
+0.20
162.01 -0.20
+0.20
150.00 -0.20


Ossur CTI3 Brace
Electronics Enclosure
Arduino Uno
Rotary Encoder
iPhone 14 Pro
Phone Mount

Failed Prototypes





Software and Code

Code

A separate Python code uses colour masking to isolate the specific pen marks on the patient's patella by filtering out all other background colours. As the leg moves, the encoder reads the leg flexion angle, and the Arduino sends the value to the main Python code on the laptop. The main Python code matches this data with the iPhone 3D depth data to calculate the patellar mediolateral shift throughout the knee's flexion from 0-90º. Figure 1 shows a flowchart of the code's logic.

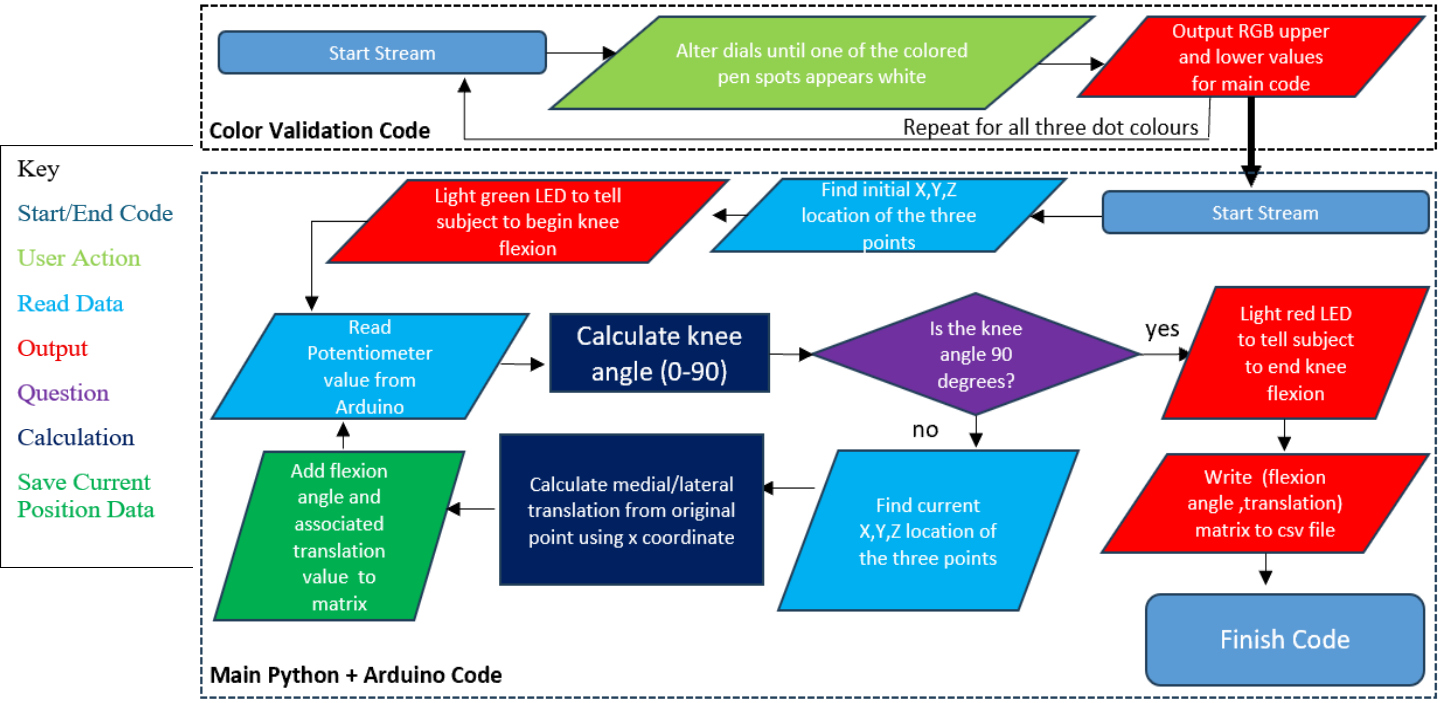


Figure 1 – Flowchart demonstrating data gathering and interpretation within the code

Final Code

```
import numpy as np
import cv2
from record3d import Record3DStream
from threading import Event


# --- MANDATORY CONSTRAINTS ---
MAX_POINTS = 30
```

```
class BifurcationTracker:
    def __init__(self):
        self.event = Event()
        self.session = None
        self.marker_ranges = []
        self.data_log = []
        self.recording = False

        # Presets for [H_min, S_min, V_min, H_max, S_max, V_max]
        self.presets = [
            [0, 100, 100, 10, 255, 255],   # 1. Red-ish
            [40, 100, 100, 80, 255, 255],  # 2. Green-ish
            [100, 100, 100, 140, 255, 255] # 3. Blue-ish
        ]

    def on_new_frame(self):
        self.event.set()

    def nothing(self, x): pass

    def calibrate_marker(self, marker_num):
        win_name = f'Calibrating Marker {marker_num}'
        cv2.namedWindow(win_name)
        p = self.presets[marker_num - 1]

        # Create sliders with the color presets
        cv2.createTrackbar('H Min', win_name, p[0], 179, self.nothing)
        cv2.createTrackbar('S Min', win_name, p[1], 255, self.nothing)
        cv2.createTrackbar('V Min', win_name, p[2], 255, self.nothing)
        cv2.createTrackbar('H Max', win_name, p[3], 179, self.nothing)
        cv2.createTrackbar('S Max', win_name, p[4], 255, self.nothing)
```

```
        cv2.createTrackbar('V Max', win_name, p[5], 255, self.nothing)

        print(f"--- Marker {marker_num} (Preset: {'Red' if marker_num==1 else 'Green' if marker_num==2 else 'Blue'}) ---")
        print("Dial in until just the sticker is white, then press ENTER.")

        while True:
            self.event.wait()
            rgb = self.session.get_rgb_frame()
            hsv = cv2.cvtColor(rgb, cv2.COLOR_RGB2HSV)

            l_h = cv2.getTrackbarPos('H Min', win_name)
            l_s = cv2.getTrackbarPos('S Min', win_name)
            l_v = cv2.getTrackbarPos('V Min', win_name)
            u_h = cv2.getTrackbarPos('H Max', win_name)
            u_s = cv2.getTrackbarPos('S Max', win_name)
            u_v = cv2.getTrackbarPos('V Max', win_name)

            lower, upper = np.array([l_h, l_s, l_v]), np.array([u_h, u_s, u_v])
            mask = cv2.inRange(hsv, lower, upper)

            cv2.imshow(win_name, mask)
            if cv2.waitKey(1) & 0xFF == 13: # ENTER
                self.marker_ranges.append((lower, upper))
                break
            self.event.clear()

        cv2.destroyWindow(win_name)

    def run(self):
        devs = Record3DStream.get_connected_devices()
```

```
self.session = Record3DStream()
self.session.on_new_frame = self.on_new_frame
self.session.connect(devs[0])

# Runs 3 times automatically
for i in range(1, 4):
    self.calibrate_marker(i)

print("\nCalibrated! Press 'r' to start recording 30 points.")

while True:
    self.event.wait()
    rgb = self.session.get_rgb_frame()
    hsv = cv2.cvtColor(rgb, cv2.COLOR_RGB2HSV)
    frame = cv2.cvtColor(rgb, cv2.COLOR_RGB2BGR)
    current_frame = []

    for low, upp in self.marker_ranges:
        mask = cv2.inRange(hsv, low, upp)
        conts, _ = cv2.findContours(mask, cv2.RETR_TREE, cv2.CHAIN_APPROX_SIMPLE)
        if conts:
            c = max(conts, key=cv2.contourArea)
            if cv2.contourArea(c) > 50:
                x, y, w, h = cv2.boundingRect(c)
                cv2.rectangle(frame, (x, y), (x+w, y+h), (0, 255, 0), 2)
                current_frame.append([x, y])

    # Record logic: Save only if we see all 3 markers
    if self.recording and len(current_frame) == 3:
        self.data_log.append(current_frame)
        if len(self.data_log) >= MAX_POINTS: # Max 30 points
```

```
                self.save_data()
                self.recording = False

            cv2.imshow('Tracking Window', frame)
            key = cv2.waitKey(1) & 0xFF
            if key == ord('q'): break
            if key == ord('r'):
                self.data_log = []
                self.recording = True
                print("Recording...")
            self.event.clear()

    def save_data(self):
        arr = np.array(self.data_log)
        # Generate the three required files
        np.savetxt("curve_1.csv", arr[:, 0, 0], delimiter=",")
        np.savetxt("curve_2.csv", arr[:, 1, 0], delimiter=",")
        np.savetxt("local_bifurcation.csv", arr.reshape(len(arr), -1), delimiter=",")
        print(f'Bifurcation capture complete. Saved {len(arr)} points to 3 files.") #

if __name__ == '__main__':
    BifurcationTracker().run()
```

Encoder Code:

```
#include <math.h>

// Pins
const int PIN_LED1  = 5;
const int PIN_LED2  = 9;
const int PIN_BTN1  = 7;
```

```
const int PIN_BTN2  = 11;
const int ENC_A     = 2;
const int ENC_B     = 3;
const int PPR       = 128;

//  Encoder
volatile long encCount = 0;

void encoderISR() {
  encCount += (digitalRead(ENC_B) == LOW) ? 1 : -1;
}

long safeCount() {
  noInterrupts();
  long c = encCount;
  interrupts();
  return c;
}

float getAngle() {
  float a = fmod(((float)safeCount() / PPR) * 360.0, 360.0);
  return (a < 0) ? a + 360.0 : a;
}

//  Calibration zone
bool inCalibZone(float a) {
  if (a >= 358.0 || a <= 4.0)  return true;
  if (a >= 43.0  && a <= 47.0) return true;
  if (a >= 88.0  && a <= 92.0) return true;
  return false;
}
```

```
// Modes
bool calibMode  = false;
bool recordMode = false;

// Recording
unsigned long recStart      = 0;
bool nearTarget[3]          = {false, false, false};
const float TARGETS[3]      = {0.0, 45.0, 90.0};
const float TOL             = 1.5;

bool isNearTarget(float a, int i) {
  float t = TARGETS[i];
  if (t == 0.0) return (a <= TOL || a >= 360.0 - TOL);
  return fabs(a - t) <= TOL;
}

// ── Flash helper (blocking)
void flashBoth(int times, unsigned long ms) {
  for (int i = 0; i < times; i++) {
    digitalWrite(PIN_LED1, HIGH);
    digitalWrite(PIN_LED2, HIGH);
    delay(ms);
    digitalWrite(PIN_LED1, LOW);
    digitalWrite(PIN_LED2, LOW);
    delay(ms);
  }
}

// ── Button debounce
```

```
const unsigned long DEB     = 50;
const unsigned long DBL_WIN = 350;

int  b1Raw = HIGH, b1Stable = HIGH;
unsigned long b1DebTimer  = 0;
int  b1Clicks             = 0;
unsigned long b1LastClick = 0;

int  b2Raw = HIGH, b2Stable = HIGH;
unsigned long b2DebTimer  = 0;

bool debouncePress(int pin, int &raw, int &stable, unsigned long &timer) {
  int r = digitalRead(pin);
  if (r != raw) { timer = millis(); raw = r; }
  if ((millis() - timer > DEB) && (r != stable)) {
    stable = r;
    return (stable == LOW);
  }
  return false;
}

void flushButtons() {
  b1Raw = digitalRead(PIN_BTN1); b1Stable = b1Raw;
  b2Raw = digitalRead(PIN_BTN2); b2Stable = b2Raw;
  b1Clicks = 0;
}

// ── Setup
void setup() {
  pinMode(PIN_LED1, OUTPUT);
```

```
  pinMode(PIN_LED2, OUTPUT);
  pinMode(PIN_BTN1, INPUT_PULLUP);
  pinMode(PIN_BTN2, INPUT_PULLUP);
  pinMode(ENC_A, INPUT);
  pinMode(ENC_B, INPUT);
  attachInterrupt(digitalPinToInterrupt(ENC_A), encoderISR, RISING);
  Serial.begin(9600);
}

// ── Loop
void loop() {
  unsigned long now = millis();
  float angle = getAngle();

  // ── Button 1: accumulate clicks
  if (debouncePress(PIN_BTN1, b1Raw, b1Stable, b1DebTimer)) {
    b1Clicks++;
    b1LastClick = now;
  }

  // ── Button 1: fire after double-press window expires
  if (b1Clicks > 0 && (now - b1LastClick) > DBL_WIN) {
    int clicks = b1Clicks;
    b1Clicks = 0;

    if (clicks >= 2) {
      // Double press: zero encoder
      long snap;
      noInterrupts();
      snap     = encCount;
```

```
    encCount = 0;
    interrupts();

    Serial.print("encoder count: ");
    Serial.print(snap);
    Serial.println(", angle zeroed");

    unsigned long end = millis() + 2000;
    while (millis() < end) {
      digitalWrite(PIN_LED1, HIGH); delay(100);
      digitalWrite(PIN_LED1, LOW);  delay(100);
    }
    flushButtons();
    if (recordMode) digitalWrite(PIN_LED2, HIGH);

  } else {
    // Single press: toggle calibration
    if (!calibMode) {
      calibMode = true;
      Serial.println("calibration mode");
      flashBoth(2, 500);
    } else {
      calibMode = false;
      digitalWrite(PIN_LED1, LOW);
      flashBoth(3, 500);
      Serial.println("calibration mode exited");
    }
    flushButtons();
    if (recordMode) digitalWrite(PIN_LED2, HIGH);
  }
}
```

```
// ── Button 2: toggle recording
if (debouncePress(PIN_BTN2, b2Raw, b2Stable, b2DebTimer)) {
  if (!recordMode) {
    recordMode = true;
    recStart   = millis();
    for (int i = 0; i < 3; i++) nearTarget[i] = isNearTarget(angle, i);
    digitalWrite(PIN_LED2, HIGH);
    Serial.println("Recording");
  } else {
    recordMode = false;
    digitalWrite(PIN_LED2, LOW);
    Serial.println("recording finished");
  }
}

// ── Calibration: PIN_LED1 tracks angle zones
if (calibMode) {
  digitalWrite(PIN_LED1, inCalibZone(angle) ? HIGH : LOW);
}

// ── Recording: print [time, angle] on target crossing
if (recordMode) {
  for (int i = 0; i < 3; i++) {
    bool near = isNearTarget(angle, i);
    if (near && !nearTarget[i]) {
      Serial.print("[");
      Serial.print(millis() - recStart);
      Serial.print(", ");
      Serial.print((int)TARGETS[i]);
```

```
        Serial.println("]");
      }
      nearTarget[i] = near;
    }
  }
}
```

Relevant International Standards

List of Applicable Standards

1. International Standards Office. ISO 20685-1:2018. *3D Scanning Methodologies for Internationally Compatible Anthropometric Databases.* Geneva: ISO, 2018.
2. International Standards Office. ISO 9241-210:2019. *Ergonomics of Human-System Interaction.* Geneva: ISO, 2019.
3. British Standards Institution. BS EN ISO 17664-1:2021. *Processing of Healthcare Products.* London: BSI, 2021.
4. British Standards Institution. BS EN ISO 7250:2017. *Basic Human Body Measurements for Technological Design*. London: BSI, 2017.
5. British Standards Institution. BS EN ISO 10993:2025. *Biological Evaluation of Medical Devices*. London: BSI, 2025.
6. British Standards Institution. BS EN ISO 10328:2016. *Structural Testing of Lower Limb Prostheses.* London: BSI, 2016
7. International Standards Office. ISO 6385:2016. *Ergonomics Principles in the Design of Work Systems*. Geneva: ISO, 2016.
8. International Standards Office. ISO 13485:2016. *Medical Devices – Quality Management Systems.* Geneva: ISO, 2016.
9. International Electrotechnical Commission. PD IEC/TR 62366:2016. *Medical Devices – Guidance on the Application of Usability Engineering to Medical Devices.* Geneva: IEC, 2016.
10. International Standards Office. ISO 15223-1:2021. *Medical Devices – Symbols to be Supplied by the Manufacturer.* Geneva: ISO, 2021.
11. International Standards Office. ISO 62304:2006. *Medical Device Software*. Geneva: ISO, 2006.
12. International Electrotechnical Commission. IEC 60601-1:2015. *Medical Electrical Equipment.* Geneva: IEC, 2015.
13. International Standards Office. ISO 14971:2019. *Medical Devices – Application of Risk Management to Medical Devices.* Geneva: ISO, 2019.
14. International Electrotechnical Commission. IEC 62133:2017. *Safety Testing for Lithium Ion Batteries.* Geneva: IEC, 2017.
15. British Standards Institution. BS EN ISO 15883-1:2025. *Washer-Disinfectors.* London: BSI, 2025.
16. International Standards Office. ISO 14937:2009. *Sterilisation of Healthcare Products*. Geneva: ISO, 2009.

Material and Component Data Sheets

Fuse Deposition Modelled Polylactic Acid (FDM PLA)

Technical data sheet PLA

Ultimaker

Chemical composition	See PLA safety data sheet, section 3
Description	Ultimaker PLA filament provides a no-hassle 3D printing experience thanks to its reliability and good surface quality. Our PLA is made from organic and renewable sources. It's safe, easy to print with, and it serves a wide range of applications for both novice and advanced users
Key features	Good tensile strength and surface quality, easy to work with at high print speeds, user-friendly for both home and office environments, PLA allows the creation of high-resolution parts. There is a wide range of color options available
Applications	Household tools, toys, educational projects, show objects, prototyping, architectural models, as well as lost casting methods to create metal parts
Non-suitable for	Food contact and *in vivo* applications. Long term outdoor usage or applications where the printed part is exposed to temperatures higher than 50 °C

Filament specifications

	Value	Method
Diameter	2.85 ± 0.10 mm	-
Max roundness deviation	0.10 mm	-
Net filament weight	350 g / 750 g	-
Filament length	~ 44 m / ~ 95 m	-

Color information

Color	Color code
PLA Green	RAL 6018
PLA Black	RAL 9005
PLA Silver Metallic	RAL 9006
PLA White	RAL 9010
PLA Transparant	N/A
PLA Orange	RAL 2008
PLA Blue	RAL 5002
PLA Magenta	RAL 4010
PLA Red	RAL 3020
PLA Yellow	RAL 1003
PLA Pearl White	RAL 1013

Mechanical properties*

	Injection molding		3D printing	
	Typical value	**Test method**	**Typical value**	**Test method**
Tensile modulus	-	-	2,346.5 MPa	ISO 527 (1 mm/min)
Tensile stress at yield	-	-	49.5 MPa	ISO 527 (50 mm/min)
Tensile stress at break	-	-	45.6 MPa	ISO 527 (50 mm/min)
Elongation at yield	-	-	3.3%	ISO 527 (50 mm/min)
Elongation at break	-	-	5.2%	ISO 527 (50 mm/min)
Flexural strength	-	-	103 MPa	ISO 178
Flexural modulus	-	-	3,150 MPa	ISO 178
Izod impact strength, notched (at 23 °C)	-	-	5.1 kJ/m²	ISO 180
Charpy impact strength (at 23 °C)	-	-	-	
Hardness	-	-	83 (Shore D)	Durometer

Electrical properties*

	Typical value	**Test method**	**Typical value**	**Test method**
Dissipation factor (at 1 MHz)	-	-	0.008	ASTM D150-11
Dielectric constant (at 1 MHz)	-	-	2.70	ASTM D150-11

Thermal properties

	Typical value	**Test method**
Melt mass-flow rate (MFR)	6.09 g/10 min	ISO 1133 (210 °C, 2.16 kg)
Heat detection (at 0.455 MPa)	-	-
Heat deflection (at 1.82 MPa)	-	-
Vicat softening temperature	-	-
Glass transition	~ 60 °C	ISO 11357
Coefficient of thermal expansion	-	-
Melting temperature	145 - 160 °C	ISO 11357
Thermal shrinkage	-	-

*See notes

Other properties

	Value	Test method
Specific gravity	1.24	ASTM D1505
Flame classification	-	-

Notes

Properties reported here are average of a typical batch. The 3D printed test specimens were printed in the XY plane, using the normal quality profile in Ultimaker Cura 2.1, an Ultimaker 2+, a 0.4 mm nozzle, 90% infill, 210 °C nozzle temperature, and 60 °C. The values are the average of five white and five black specimens for the tensile, flexural, and impact tests. The Shore hardness D was measured in a 7-mm-thick square printed using the normal quality profile in Utlimaker Cura 2.5, an Ultimaker 3, a 0.4 mm print core, and 100% infill. The electrical properties were measured on a 54-mm-diameter disk with 3 mm thickness printed in the XY plane, using the fine quality profile (0.1 mm layer height) in Ultimaker Cura 3.2.1, an Ultimaker 3, a 0.4 mm print core, and 100% infill. Ultimaker is constantly working on extending the TDS data.

Disclaimer

Any technical information or assistance provided herein is given and accepted at your risk, and neither Ultimaker nor its affiliates make any warranty relating to it or because of it. Neither Ultimaker nor its affiliates shall be responsible for the use of this information, or of any product, method or apparatus mentioned, and you must make your own determination of its suitability and completeness for your own use, for the protection of the environment, and for the health and safety of your employees and purchasers of your products. No warranty is made of the merchantability or fitness of any product; and nothing herein waives any of Ultimaker's conditions of sale. Specifications are subject to change without notice.

Version	Version 4.002
Date	November 19, 2018

3DOLOGiE

To purchase visit www.3DOLOGIE.com | 877.876.8372

EZ Removable Ink® DERMarker®

VISCOT®
MEDICAL, LLC.



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VLIT-52 Ver.1

EN ENGLISH
INSTRUCTION FOR USE

1. Use an alcohol swab to remove the natural oils of the skin; allow the area to dry.
2. Mark the anatomy as required and allow markings to dry.
3. If injecting: Inject directly next to the marked spot.
4. Post procedure, use an alcohol swab with limited pressure for 30 seconds to remove marks.

INTENDED PURPOSE: Marking anatomical sites on intact skin for consultations and prior to non-surgical procedures.

INTENDED USER: To be used by clinicians on patients.

PERFORMANCE CHARACTERISTICS: Clear visible mark.

PRECAUTIONS

- Designed for non-surgical procedures only.
- Marker is for External Use, not internal use.
- Not for ocular use
- Not for use on newborns
- Reuse of single use device could result in infection/ contamination and/or device failure which could lead to patient harm.

HANDLING: No special requirements needed

NOTICE:
Any serious incident that occurs with the device should be reported to the manufacturer and the Competent Authority of the Member State in which the user/patient is established.

Medical Grade Pen

*

1 Identification

· Product identifier

· Trade name: Easily Removable Ink (Green)

· Article number: 100000000505

· Relevant identified uses of the substance or mixture and uses advised against
Currently no such applications are identified

· Application of the substance / the mixture Easily Removable Ink

· Details of the supplier of the safety data sheet

· Manufacturer/Supplier:
Viscot Medical, LLC.
32 West Street
East Hanover, NJ 07936
Phone: 973-887-9273
Fax: 973-887-3961
www. viscot.com

· Information department:
Tel.: 973-887-9273

· Emergency telephone numbe
973-887-9273

*

2 Hazard(s) identification

· Classification of the substance or mixture



GHS02 Flame

Flam. Liq. 2 H225 Highly flammable liquid and vapor.



GHS07

Eye Irrit. 2A H319 Causes serious eye irritation.
STOT SE 3 H336 May cause drowsiness or dizziness.

· Label elements

· GHS label elements
The product is classified and labeled according to the Globally Harmonized System (GHS).

· Hazard pictograms



GHS02 GHS07

· Signal word Danger

· Hazard-determining components of labeling:
1-methoxy-2-propanol
propan-2-ol

· Hazard statements
Highly flammable liquid and vapor.
Causes serious eye irritation.
May cause drowsiness or dizziness.

· Precautionary statements
Keep away from heat/sparks/open flames/hot surfaces. - No smoking.
Use explosion-proof electrical/ventilating/lighting/equipment.
Avoid breathing dust/fume/gas/mist/vapors/spray
Wear protective gloves / eye protection / face protection.
Wear eye protection / face protection.
Ground/bond container and receiving equipment.
Use only non-sparking tools.
Take precautionary measures against static discharge.
Wash thoroughly after handling.
Use only outdoors or in a well-ventilated area.
IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/ shower.
If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
Call a poison center/doctor if you feel unwell.

(Contd. on page 2)

Trade name: Easily Removable Ink (Green)

(Contd. of page 1)

If eye irritation persists: Get medical advice/attention.
In case of fire: Use for extinction: CO2, powder or water spray.
Store locked up.
Store in a well-ventilated place. Keep container tightly closed.
Store in a well-ventilated place. Keep cool.
Dispose of contents/container in accordance with local/regional/national/international regulations.

· Classification system:
· NFPA ratings (scale 0 - 4)



Health = 1
Fire = 3
Reactivity = 0

· HMIS-ratings (scale 0 - 4)



Health = 1
Fire = 3
Reactivity = 0

· Other hazards
· Results of PBT and vPvB assessment
· PBT: Not applicable.
· vPvB: Not applicable.

*

3 Composition/information on ingredients

· Chemical characterization: Mixtures
Mixture of the following substances, containing non-hazardous substances and colouring agents.
· Description: Mixture of the substances listed below with nonhazardous additions.

· Dangerous components:

64-17-5	ethanol	50-100%
107-98-2	1-methoxy-2-propanol	10-25%
67-63-0	propan-2-ol	2.5-10%

· Additional information: For the wording of the listed risk phrases refer to section 16.

*

4 First-aid measures

· Description of first aid measures
· After inhalation: Supply fresh air; consult doctor in case of complaints.
· After skin contact: Generally the product does not irritate the skin.
· After eye contact:
Rinse opened eye for several minutes under running water. If symptoms persist, consult a doctor.
· After swallowing: If symptoms persist consult doctor.
· Information for doctor:
· Most important symptoms and effects, both acute and delayed
No further relevant information available.
· Indication of any immediate medical attention and special treatment needed
No further relevant information available.

*

5 Fire-fighting measures

· Extinguishing media
· Suitable extinguishing agents:
CO2, extinguishing powder or water spray. Fight larger fires with water spray or alcohol resistant foam.
· Special hazards arising from the substance or mixture No further relevant information available.
· Advice for firefighters
· Protective equipment: No special measures required.

*

6 Accidental release measures

· Personal precautions, protective equipment and emergency procedures
Wear protective equipment. Keep unprotected persons away.
Wear protective clothing.
· Environmental precautions:
Prevent seepage into sewage system, workpits and cellars.
Dilute with plenty of water.
Do not allow to enter sewers/ surface or ground water.
· Methods and material for containment and cleaning up:
Absorb with liquid-binding material (sand, diatomite, acid binders, universal binders, sawdust).

(Contd. on page 3)

(Contd. of page 2)

Ensure adequate ventilation.
· Reference to other sections
See Section 7 for information on safe handling.
See Section 8 for information on personal protection equipment.
See Section 13 for disposal information.

*
7 Handling and storage

· Handling:
· Precautions for safe handling
Ensure good ventilation/exhaustion at the workplace.
Prevent formation of aerosols.
· Information about protection against explosions and fires:
Keep ignition sources away - Do not smoke.
Protect from heat.
Protect against electrostatic charges.

· Conditions for safe storage, including any incompatibilities
· Storage:
· Requirements to be met by storerooms and receptacles: Store in a cool location.
· Information about storage in one common storage facility: Not required.
· Further information about storage conditions:
Keep receptacle tightly sealed.
Store in cool, dry conditions in well sealed receptacles.
Protect from heat and direct sunlight.
· Specific end use(s) No further relevant information available.

*
8 Exposure controls/personal protection

· Additional information about design of technical systems: No further data; see item 7.

· Control parameters

· Components with limit values that require monitoring at the workplace:	
64-17-5 ethanol (50-100%)	
PEL	Long-term value: 1900 mg/m³, 1000 ppm
REL	Long-term value: 1900 mg/m³, 1000 ppm
TLV	Short-term value: 1880 mg/m³, 1000 ppm
107-98-2 1-methoxy-2-propanol (10-25%)	
REL	Short-term value: 540 mg/m³, 150 ppm Long-term value: 360 mg/m³, 100 ppm
TLV	Short-term value: 369 mg/m³, 100 ppm Long-term value: 184 mg/m³, 50 ppm
67-63-0 propan-2-ol (2.5-10%)	
PEL	Long-term value: 980 mg/m³, 400 ppm
REL	Short-term value: 1225 mg/m³, 500 ppm Long-term value: 980 mg/m³, 400 ppm
TLV	Short-term value: 984 mg/m³, 400 ppm Long-term value: 492 mg/m³, 200 ppm BEI

· Ingredients with biological limit values:	
67-63-0 propan-2-ol (2.5-10%)	
BEI	40 mg/L Medium: urine Time: end of shift at end of workweek Parameter: Acetone (background, nonspecific)

· Additional information: The lists that were valid during the creation were used as basis.

· Exposure controls
· Personal protective equipment:
· General protective and hygienic measures:
Keep away from foodstuffs, beverages and feed.
Immediately remove all soiled and contaminated clothing.
Wash hands before breaks and at the end of work.
Avoid contact with the eyes.
Avoid contact with the eyes and skin.
· Breathing equipment:
In case of brief exposure or low pollution use respiratory filter device. In case of intensive or longer exposure use respiratory protective device that is independent of circulating air.
· Protection of hands:
The glove material has to be impermeable and resistant to the product/ the substance/ the preparation.

(Contd. on page 4)

(Contd. of page 3)

Due to missing tests no recommendation to the glove material can be given for the product/ the preparation/ the chemical mixture.
Selection of the glove material on consideration of the penetration times, rates of diffusion and the degradation

· Material of gloves
The selection of the suitable gloves does not only depend on the material, but also on further marks of quality and varies from manufacturer to manufacturer. As the product is a preparation of several substances, the resistance of the glove material can not be calculated in advance and has therefore to be checked prior to the application.

· Penetration time of glove material
The exact break trough time has to be found out by the manufacturer of the protective gloves and has to be observed.

· Eye protection:



Tightly sealed goggles

*

9 Physical and chemical properties

· Information on basic physical and chemical properties	
· General Information	
· Appearance:	
Form:	Fluid
Color:	According to product specification
· Odor:	Alcohol-like
· Odour threshold:	Not determined.
· Important information on protection of health and environment, and on safety.	-
· pH-value:	Not determined.
· Change in condition	
Melting point/Melting range:	Undetermined.
Boiling point/Boiling range:	78 °C (172 °F)
· Flash point:	13 °C (55 °F)
· Flammability (solid, gaseous):	Not applicable.
· Ignition temperature:	287 °C (549 °F)
· Decomposition temperature:	Not determined.
· Auto igniting:	Product is not selfigniting.
· Danger of explosion:	Product is not explosive. However, formation of explosive air/vapor mixtures are possible.
· Explosion limits:	
Lower:	1.7 Vol %
Upper:	15.0 Vol %
· Vapor pressure at 20 °C (68 °F):	59 hPa (44 mm Hg)
· Density:	Not determined.
· Relative density	Not determined.
· Vapour density	Not determined.
· Evaporation rate	Not determined.
· Solubility in / Miscibility with	
Water:	Partly miscible.
· Partition coefficient (n-octanol/water):	Not determined.
· Viscosity:	
Dynamic:	Not determined.
Kinematic:	Not determined.
· Solvent content:	
Organic solvents:	81.6 %
Solids content:	7.4 %
· Other information	The physical and chemical properties given in Section 9.1 are rough data only, which are partially derived from the component's data of the mixture. These data are no binding product specifications.

US

(Contd. on page 5)

Trade name: Easily Removable Ink (Green)

(Contd. of page 4)

*

10 Stability and reactivity

· Reactivity
· Chemical stability
· Thermal decomposition / conditions to be avoided:
No decomposition if used according to specifications.
· Possibility of hazardous reactions No dangerous reactions known.
· Conditions to avoid No further relevant information available.
· Incompatible materials: No further relevant information available.
· Hazardous decomposition products: No dangerous decomposition products known.

*

11 Toxicological information

· Information on toxicological effects
· Acute toxicity:
· Primary irritant effect:
· on the skin: No irritant effect.
· on the eye: Irritating effect.
· Sensitization: No sensitizing effects known.
· Additional toxicological information:
The product shows the following dangers according to internally approved calculation methods for preparations:
Irritant

· Carcinogenic categories

· IARC (International Agency for Research on Cancer)	
67-63-0 propan-2-ol	3

· NTP (National Toxicology Program)
None of the ingredients is listed.

· OSHA-Ca (Occupational Safety & Health Administration)
None of the ingredients is listed.

*

12 Ecological information

· Toxicity
· Aquatic toxicity: No further relevant information available.
· Persistence and degradability No further relevant information available.
· Behavior in environmental systems:
· Bioaccumulative potential No further relevant information available.
· Mobility in soil No further relevant information available.
· Additional ecological information:
· General notes:
Water hazard class 1 (Self-assessment): slightly hazardous for water
Do not allow undiluted product or large quantities of it to reach ground water, water course or sewage system.
· Results of PBT and vPvB assessment
· PBT: Not applicable.
· vPvB: Not applicable.
· Other adverse effects No further relevant information available.

*

13 Disposal considerations

· Waste treatment methods
· Recommendation:
Must not be disposed of together with household garbage. Do not allow product to reach sewage system.

· Uncleaned packagings:
· Recommendation: Disposal must be made according to official regulations.
· Recommended cleansing agent: Water, if necessary with cleansing agents.

*

14 Transport information

· UN-Number	
· DOT, ADR, IMDG, IATA	UN1263
· UN proper shipping name	
· DOT	Paint
· ADR	1263 Paint
· IMDG, IATA	PAINT

(Contd. on page 6)

Trade name: Easily Removable Ink (Green)

(Contd. of page 5)

· Transport hazard class(es)

· DOT



· Class	3 Flammable liquids
· Label	3

· ADR



· Class	3 (F1) Flammable liquids
· Label	3

· IMDG, IATA



· Class	3 Flammable liquids
· Label	3
· Packing group	
· DOT, ADR, IMDG, IATA	II
· Environmental hazards:	
· Marine pollutant:	No
· Special precautions for user	Warning: Flammable liquids
· Danger code (Kemler):	33
· EMS Number:	F-E,S-E___
· Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code	Not applicable.
· Transport/Additional information:	
· DOT	
· Quantity limitations	On passenger aircraft/rail: 5 L On cargo aircraft only: 60 L
· ADR	
· Excepted quantities (EQ)	Code: E2 Maximum net quantity per inner packaging: 30 ml Maximum net quantity per outer packaging: 500 ml
· IMDG	
· Limited quantities (LQ)	5L
· Excepted quantities (EQ)	Code: E2 Maximum net quantity per inner packaging: 30 ml Maximum net quantity per outer packaging: 500 ml
· UN "Model Regulation":	UN1263, Paint, 3, II

*

15 Regulatory information

· Safety, health and environmental regulations/legislation specific for the substance or mixture

· Sara

· Section 355 (extremely hazardous substances):

None of the ingredient is listed.

· Section 313 (Specific toxic chemical listings):

None of the ingredients is listed.

· TSCA (Toxic Substances Control Act):

All ingredients are listed.

· Proposition 65

· Chemicals known to cause cancer:

None of the ingredients is listed.

· Chemicals known to cause reproductive toxicity for females:

None of the ingredients is listed.

(Contd. on page 7)

Trade name: Easily Removable Ink (Green)

(Contd. of page 6)

· Chemicals known to cause reproductive toxicity for males:
None of the ingredients is listed.

· Chemicals known to cause developmental toxicity:
64-17-5 ethanol

· DSL/NDSL (Canada) All ingredients are listed

· Cancerogenity categories

· EPA (Environmental Protection Agency)
None of the ingredients is listed.

· TLV (Threshold Limit Value established by ACGIH)

64-17-5	ethanol	A3
67-63-0	propan-2-ol	A4

· MAK (German Maximum Workplace Concentration)

64-17-5	ethanol	5

· NIOSH-Ca (National Institute for Occupational Safety and Health)
None of the ingredients is listed.

· GHS label elements
The product is classified and labeled according to the Globally Harmonized System (GHS).

· Hazard pictograms



GHS02 GHS07

· Signal word Danger

· Hazard-determining components of labeling:
1-methoxy-2-propanol
propan-2-ol

· Hazard statements
Highly flammable liquid and vapor.
Causes serious eye irritation.
May cause drowsiness or dizziness.

· Precautionary statements
Keep away from heat/sparks/open flames/hot surfaces. - No smoking.
Use explosion-proof electrical/ventilating/lighting/equipment.
Avoid breathing dust/fume/gas/mist/vapors/spray
Wear protective gloves / eye protection / face protection.
Wear eye protection / face protection.
Ground/bond container and receiving equipment.
Use only non-sparking tools.
Take precautionary measures against static discharge.
Wash thoroughly after handling.
Use only outdoors or in a well-ventilated area.
IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/ shower.
If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
Call a poison center/doctor if you feel unwell.
If eye irritation persists: Get medical advice/attention.
In case of fire: Use for extinction: CO2, powder or water spray.
Store locked up.
Store in a well-ventilated place. Keep container tightly closed.
Store in a well-ventilated place. Keep cool.
Dispose of contents/container in accordance with local/regional/national/international regulations.

· National regulations:

· Technical instructions (air):

Class	Share in %
NK	50-100

· Water hazard class: Water hazard class 1 (Self-assessment): slightly hazardous for water.
· Chemical safety assessment: A Chemical Safety Assessment has not been carried out.

16 Other information

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

· Date of preparation / last revision 05/26/2015 / 19
· Abbreviations and acronyms:
RID: Règlement international concernant le transport des marchandises dangereuses par chemin de fer (Regulations Concerning the International Transport of Dangerous Goods by Rail)

(Contd. on page 8)

Rotary Encoder



Features

- Two channel quadrature output
- Bushing or servo mount
- Square wave signal
- Small size
- Resolution to 256 PPR
- CMOS and TTL compatible
- Long life
- Ball bearing option for high operating speed up to 3000 rpm
- RoHS compliant*

EN - Rotary Optical Encoder

General Information

The Bourns® EN model is a self-contained rotary optical encoder. It produces a 2-bit quadrature signal which is suitable for digital systems where both magnitude and direction of adjustment must be provided. The EN encoder is ideal for use as a digital panel control or as a position sensing device in applications where long life, reliability, high resolution and precise linearity are critical.

The EN series encoder converts rotary input into electrical signals which can be used by microprocessors without A/D conversion.

Bourns encoder output signals are square wave digital pulses which do not require debounce circuitry. Both features make it possible to significantly reduce the memory overhead, wiring and wiring interconnects required by other types of control devices.

EN optical encoders offer a useful rotational life of from 10 million to 200 million shaft revolutions, making them ideal for extended service applications. The Bourns encoder is also compact and well suited for situations where the available space is limited.

Additional Information

Click these links for more information:

PRODUCT SELECTOR | TECHNICAL LIBRARY | INVENTORY | SAMPLES | CONTACT

Electrical Characteristics

Output	2-bit quadrature code, Channel A leads Channel B by 90 ° (electrical) with clockwise rotation
Resolution	25 to 256 cycles per revolution
Insulation Resistance (500 VDC)	1,000 megohms
Electrical Travel	Continuous
Supply Voltage	5.0 VDC ±0.25 VDC
Supply Current	26 mA maximum
Output Voltage	
Low Output	0.8 V maximum
High Output	4 V minimum
Output Current	
Low Output	25 mA minimum
Rise/Fall Time	200 ns (typical)
Shaft RPM (Ball Bearing)	3,000 rpm maximum
Power Consumption	136 mW maximum
Pulse Width (Electrical Degrees, Each Channel)	180 ° ±45 ° typ.
Pulse Width (Index Channel)	360 ° ±90 °
Phase (Electrical Degrees, Channel A to Channel B)	90 ° ±45 ° typ.

Environmental Characteristics

Operating Temperature Range	-40 °C to +75 °C (-40 °F to +167 °F)
Storage Temperature Range	-40 °C to +85 °C (-40 °F to +185 °F)
Humidity	MIL-STD-202, Method 103B, Condition B
Vibration	5 G
Shock	50 G
Rotational Life	
A & C Bushings (300 rpm maximum)**	10,000,000 revolutions
W, S & T Bushings (3,000 rpm maximum)**	200,000,000 revolutions
IP Rating	IP 40

**For resolutions ≤ 128 quadrature cycles per shaft revolution.

BOURNS®

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WARNING
Cancer and Reproductive Harm
www.P65Warnings.ca.gov

*RoHS Directive 2015/863, Mar 31, 2015 and Annex.

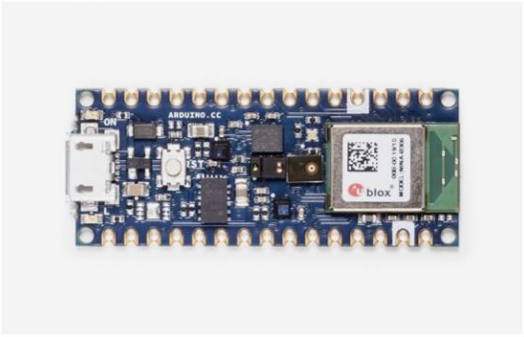
Specifications are subject to change without notice.
Users should verify actual device performance in their specific applications.
The products described herein and this document are subject to specific legal disclaimers as set forth on the last page of this document, and at www.bourns.com/docs/legal/disclaimer.pdf.

Mechanical Characteristics

Mechanical Angle	360 ° Continuous
Torque (Starting and Running)	
A & C Bushings (Spring Loaded for Optimum Feel)	1 N-cm (1.5 oz-in.) maximum
W, S & T Bushings (Ball Bearing Shaft Support)	0.07 N-cm (0.1 oz-in.) maximum
Mounting Torque	1.7 to 2.0 N-m (15 to 18 lb.-in.) maximum
Shaft End Play	0.30 mm (0.012 ″) T.I.R. maximum
Shaft Radial Play	0.12 mm (0.005 ″) T.I.R. maximum
Weight	11 gms. (0.4 oz.)
Terminals	Axial or radial pc pins or ribbon cable
Soldering Condition	
Manual Soldering	96.5Sn/3.0Ag/0.5Cu solid wire or no-clean rosin cored wire 370 °C (700 °F) max. for 3 seconds
Wave Soldering	96.5Sn/3.0Ag/0.5Cu solder with no-clean flux 260 °C (500 °F) max. for 5 seconds
Wash processes	Not recommended
Marking	Manufacturer's trademark, name, part number, and date code.
Hardware	One lockwasher and one mounting nut supplied with each encoder, except on servo mount versions.

Arduino Nano 33 BLE Sense

User Manual
SKU: ABX00031



Description

Arduino® Nano 33 BLE Sense is a miniature sized module containing a NINA B306 module, based on Nordic nRF52480 and containing an Arm® Cortex®-M4F, a crypto chip which can securely store certificates and pre shared keys and a 9 axis IMU. The module can either be mounted as a DIP component (when mounting pin headers), or as a SMT component, directly soldering it via the castellated pads

Target areas:

Maker, enhancements, IoT application

Features

- **NINA B306 Module**
  - **Processor**
    - 64 MHz Arm® Cortex®-M4F (with FPU)
    - 1 MB Flash + 256 KB RAM
  - **Bluetooth® 5 multiprotocol radio**
    - 2 Mbps
    - CSA #2
    - Advertising Extensions
    - Long Range
    - +8 dBm TX power
    - -95 dBm sensitivity
    - 4.8 mA in TX (0 dBm)
    - 4.6 mA in RX (1 Mbps)
    - Integrated balun with 50 Ω single-ended output
    - IEEE 802.15.4 radio support
    - Thread
    - Zigbee
  - **Peripherals**
    - Full-speed 12 Mbps USB
    - NFC-A tag
    - Arm CryptoCell CC310 security subsystem
    - QSPI/SPI/TWI/I²S/PDM/QDEC
    - High speed 32 MHz SPI
    - Quad SPI interface 32 MHz
    - EasyDMA for all digital interfaces
    - 12-bit 200 ksps ADC
    - 128 bit AES/ECB/CCM/AAR co-processor
- **LSM9DS1** (9-axis IMU)
  - 3 acceleration channels, 3 angular rate channels, 3 magnetic field channels
  - ±2/±4/±8/±16 g linear acceleration full scale
  - ±4/±8/±12/±16 gauss magnetic full scale
  - ±245/±500/±2000 dps angular rate full scale
  - 16-bit data output
- **LPS22HB** (Barometer and temperature sensor)
  - 260 to 1260 hPa absolute pressure range with 24 bit precision
  - High overpressure capability: 20x full-scale
  - Embedded temperature compensation
  - 16-bit temperature data output
  - 1 Hz to 75 Hz output data rateInterrupt functions: Data Ready, FIFO flags, pressure thresholds
- **HTS221** (relative humidity sensor)
  - 0-100% relative humidity range
  - High rH sensitivity: 0.004% rH/LSB
  - Humidity accuracy: ± 3.5% rH, 20 to +80% rH
  - Temperature accuracy: ± 0.5 °C,15 to +40 °C
  - 16-bit humidity and temperature output data

- **APDS-9960** (Digital proximity, Ambient light, RGB and Gesture Sensor)
  - Ambient Light and RGB Color Sensing with UV and IR blocking filters
  - Very high sensitivity – Ideally suited for operation behind dark glass
  - Proximity Sensing with Ambient light rejection
  - Complex Gesture Sensing
- **MP34DT05** (Digital Microphone)
  - AOP = 122.5 dbSPL
  - 64 dB signal-to-noise ratio
  - Omnidirectional sensitivity
  - –26 dBFS ± 3 dB sensitivity
- **ATECC608A** (Crypto Chip)
  - Cryptographic co-processor with secure hardware based key storage
  - Protected storage for up to 16 keys, certificates or data
  - ECDH: FIPS SP800-56A Elliptic Curve Diffie-Hellman
  - NIST standard P256 elliptic curve support
  - SHA-256 & HMAC hash including off-chip context save/restore
  - AES-128 encrypt/decrypt, galois field multiply for GCM
- **MPM3610** DC-DC
  - Regulates input voltage from up to 21V with a minimum of 65% efficiency @minimum load
  - More than 85% efficiency @12V

ENGLISH

SYMBOLS

MD Medical Device

INTENDED USE

The device is intended for external support, stabilization and protection of the knee.

Indications for use

For knee conditions that may benefit from increased AP and ML stability and protection, such as injuries to the ACL, MCL, LCL, and PCL, and rotary and combined instabilities.
No known contraindications.

Warnings and Cautions:

- The use of the device may increase the risk of deep vein thrombosis and pulmonary embolism.
- Care should be taken not to overtighten the device.
- Regular healthcare professional supervision is recommended for patients with peripheral vascular disease, neuropathy, and sensitive skin.

GENERAL SAFETY INSTRUCTIONS

Read these instructions carefully before use. Keep them for future reference.
Any serious incident in relation to the device must be reported to the manufacturer and relevant authorities.
The patient should stop using the device and contact a healthcare professional:

- If there is a change or loss in device functionality, or if the device shows signs of damage or wear hindering its normal functions.
- If any pain, skin irritation, excessive pressure, or unusual reaction occurs with the use of the device.

The device is for single patient – multiple use.
This device is not guaranteed to prevent injury.

FITTING INSTRUCTIONS

While carrying out the following instructions, please refer to the overview figure for locating components mentioned in the text **(Fig. 1)**.

Device Application

Correct application is vital to the proper functioning of the device.

1. While seated, bend your leg to 90°.
2. Unfasten all four straps from the lateral/outside of the device **(Fig. 2)**.
3. Place the device on your leg so that the yellow line on the Hinge (A) is slightly above the center of your kneecap. This will allow the device to seat properly when you stand **(Fig. 3)**.
   Note: Use any combination of the Condyle Pads (B) supplied to widen or narrow the width of the device at the knee **(Fig. 6)**.
4. Attach the Lower Cruciate Strap (#1, C) (directly below the knee) first as this strap helps create the brace-to-bone contact required to help keep the device in place **(Fig. 4)**.
5. Next, snugly fasten the Bottom Strap (#2, D) **(Fig. 5)**.
6. Continue with the Upper Cruciate Strap (#3, E), located just above the knee, and then fasten the Top Strap (#4, F) **(Fig. 5)**.

Note: Overtightening the top two straps can cause the device to migrate.
Note: Control the strap length by adjusting the medial/inside hook-and-loop strap end **(Fig. 7)**.
All strap lengths can be adjusted by detaching the alligator tab at the end of the strap. Adjust the length as necessary by cutting the strap and reattach the alligator tab to the strap **(Fig. 7)**.
Note: The D-rings can be changed from internal to external D-rings or replaced if a D-ring is

broken. Remove the screw to disengage.
For maximum comfort and support, wear a knee sleeve and ensure the product fits securely.

Device Adjustments

Hinge Extension Stop Set Installation

The device comes equipped with a 5° Extension Stop (G) installed. Extension Stops for 0°, 10°, 20°, and 30° are also available. The specific stop angle is molded into each stop.

1. Remove the Condyle Pad from the Hinge **(Fig. 8)**.
2. Remove the Philips screw **(Fig. 9)**.
3. Flip the device over to see the front of the device. Bend the device slightly. Place the tip of the screwdriver under the Extension Stop **(Fig. 10)** and gently pry the stop from the Hinge.
4. Select your desired stop angle **(Fig. 11)**. Angles are marked on each stop.
5. Bend the device slightly. Insert the hooked tip of the new Extension Stop into the Hinge **(Fig. 12)** and push down on the other side until it is flush with the front of the Hinge **(Fig. 13)**.
6. Finish by reinstalling the screw and Condyle Pad **(Fig. 14)**.
7. Repeat this process for the Hinge on the other side, making sure to use the matching Extension Stop.

Hinge Flexion Stop Set Installation

Installation:

1. Select the correct angle setting. 60°, 75°, and 90° Flexion Stop options are provided in the stop kit **(Fig. 15)**.
2. Place the stop into position with the Snap **(Fig. 16 A)** aligned to the hinge location (A), then slide the Gear Jam **(Fig. 16 B)** into the hinge.
3. Hold the stop in place as shown. **(Fig. 17)**
4. Slowly flex the device while ensuring the landing shelf meets the upper frame as shown.
5. The stop will audibly click into place. **(Fig. 18)**
6. Repeat for opposite side hinge, using the same size stop.

Installation of the 0° Flexion Stop:

1. Position device into full extension **(Fig. 19)** with 0° extension stops installed **(Fig. 20)**. **Note:** Device needs to be forced into full extension by applying adequate force.
2. Place the stop into position with Gear Jam **(Fig. 16 B)** pushed into the hinge gear teeth. An audible snap will confirm the correct position **(Fig. 21)**.
3. Hold the stop in place as shown. Use thumb to push Snap **(Fig. 16 A)** into place.
4. The stop will audibly click into place **(Fig. 22)**.
5. Repeat for opposite side hinge, using the same size stop.

Removal:

1. With the device in extension, grab the tail of the stop and pull away from the frame. The Flexion Stop will snap out.
2. Repeat for opposite side hinge.

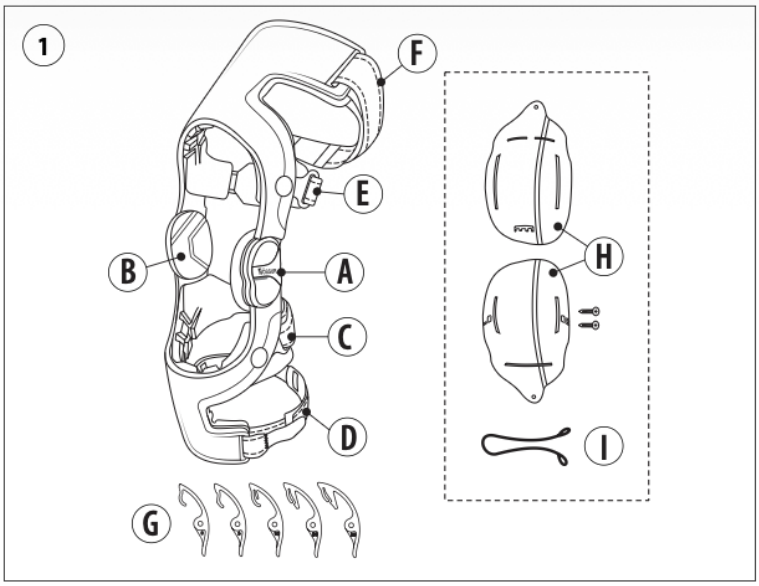
Impact Guard Set Installation - Optional Accessory

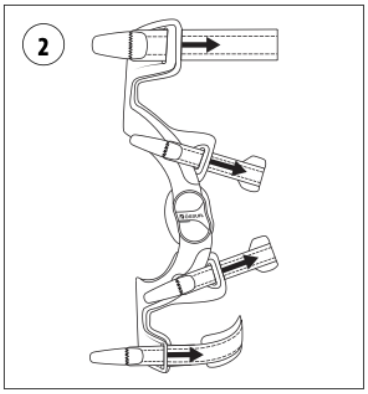
Impact Guard (H) sets are available to protect the knee against impacts. To install the Impact Guard set:

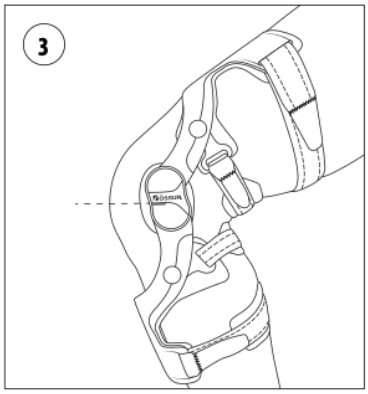
1. Carefully peel back the liners to expose the mounting tab locations on the upper and lower brace frames **(Fig. 23)**.
2. Place the Impact guard into the device, lining up the holes **(Fig. 24)**.
3. Install the two T20 Torx mounting screws to secure the Impact guard to the upper and lower brace frames **(Fig. 25)**.
4. Reinstall the upper and lower liners.

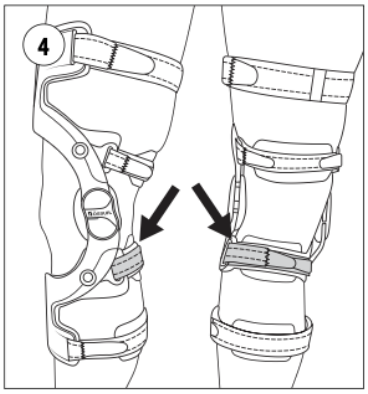
To replace the Impact guard set Elastic Cord (I) (sold separately) or reconnect the upper and lower panel of the guard set:

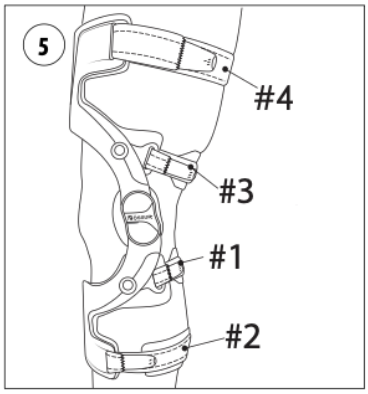
1. Fold the Elastic Cord into a loop **(Fig. 26)**.

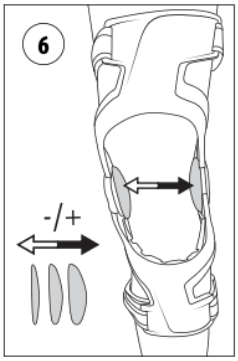

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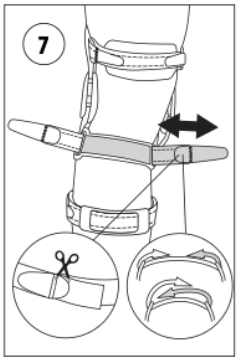

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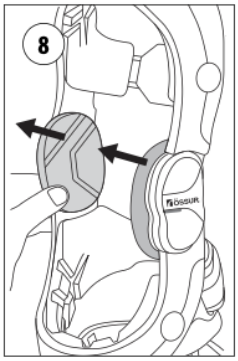

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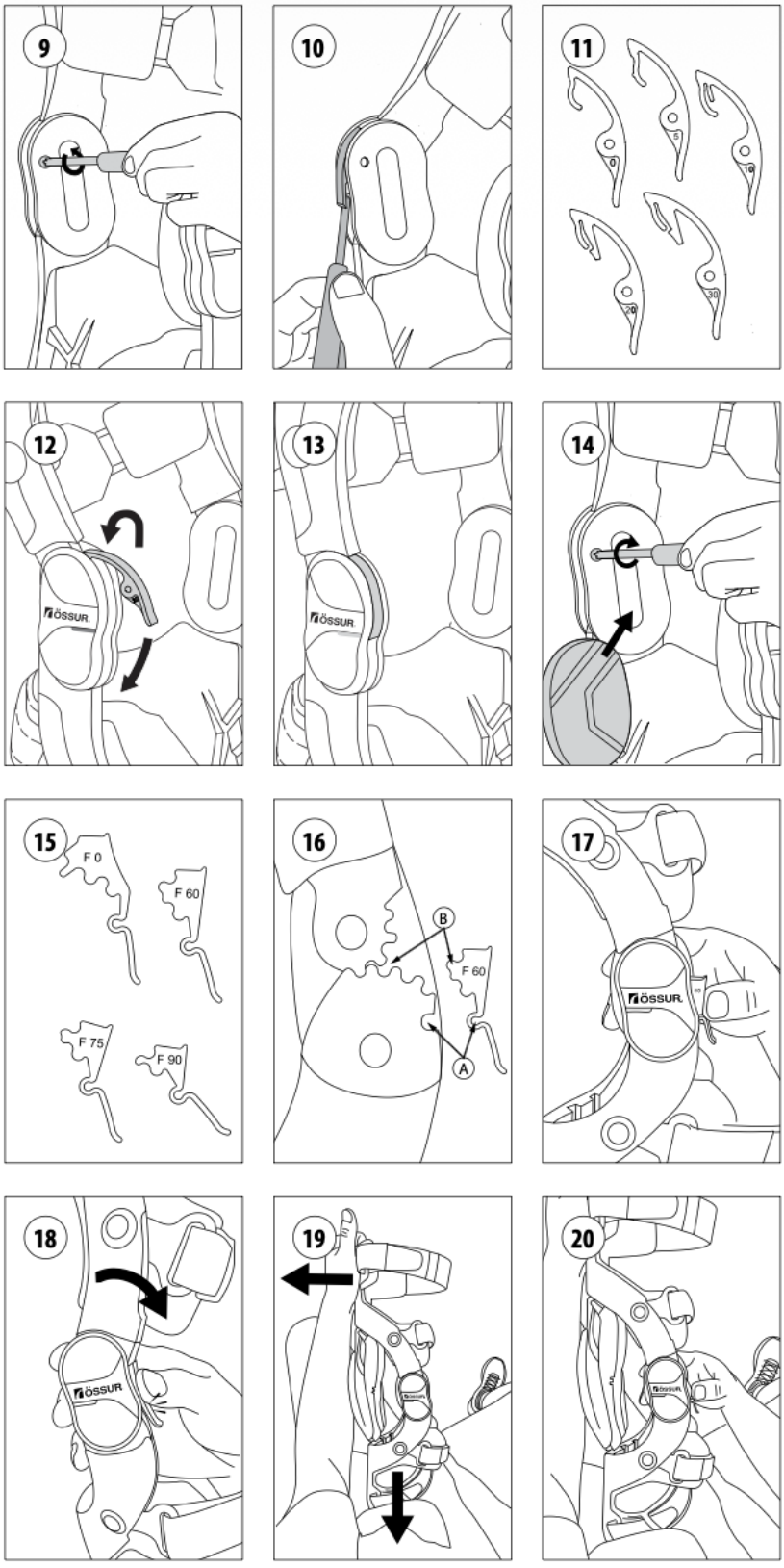

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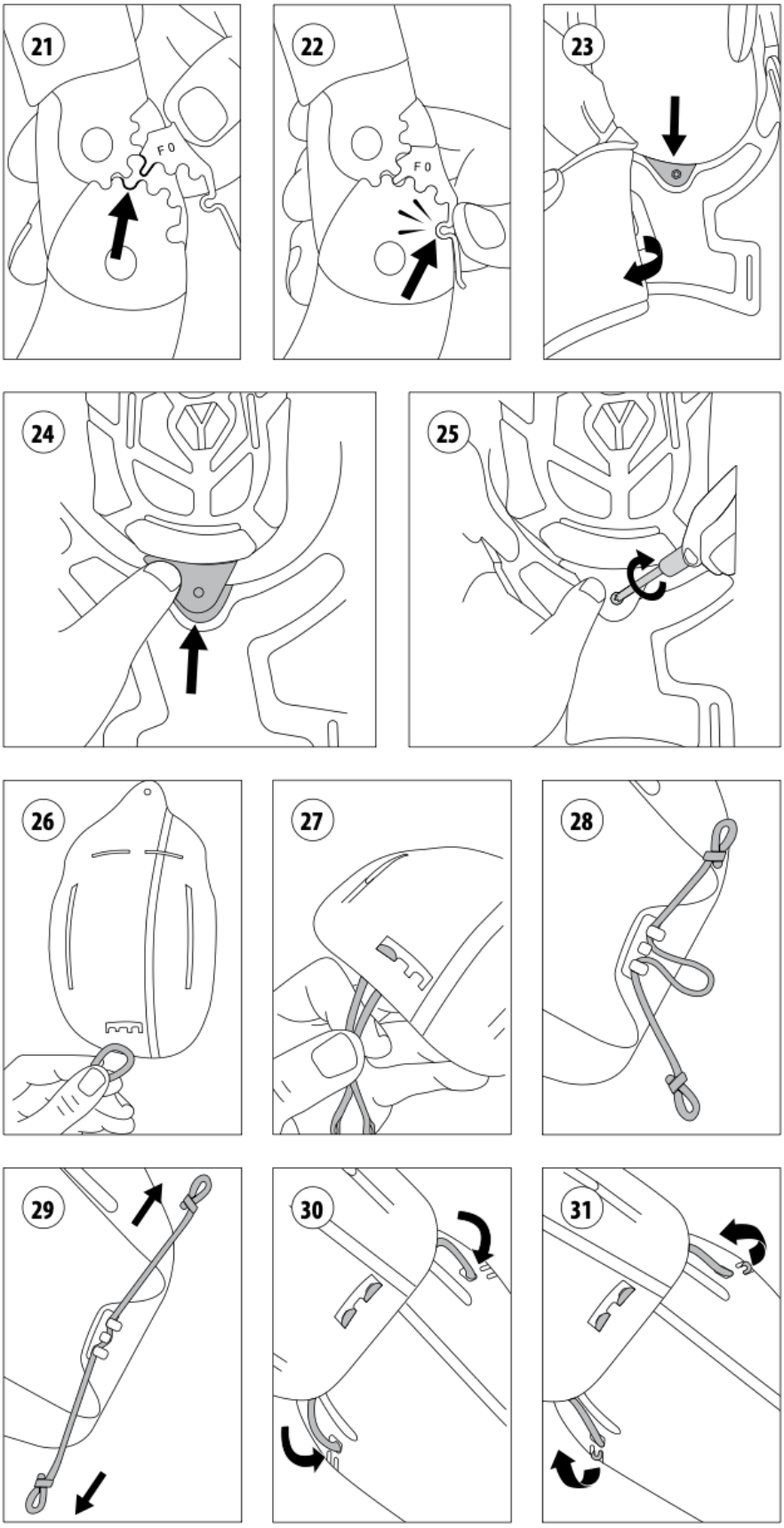

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Appendices

A1 – Meeting Log

Date: 26.01.26
In Attendance: Khaled Gadalla, Emma Graham, Daniel Kekeh
Minutes:

- Discussed scope of the project
- Decided on weekly meeting time

Date: 27.01.26
In Attendance: John Fairhurst, Khaled Gadalla, Emma Graham, Daniel Kekeh, Joshua Lipshaw
Minutes:

- Discussed individual design ideas, alongside their limitations, advantages, accuracy and ease of manufacturability
- Voted on design to go forward with – chose time of flight sensor method

Date: 30.01.26
In Attendance: John Fairhurst, Emma Graham, Daniel Kekeh, Joshua Lipshaw
Minutes:

- Composed bill of materials for design
- Discussed potential different methods of linking Arduino to remove limitations of cable length

Date: 05.02.26
In Attendance: John Fairhurst, Khaled Gadalla, Emma Graham, Daniel Kekeh, Joshua Lipshaw, Zixuan Zhang
Minutes:

- Divided into software and hardware teams
    - Hardware – CAD design, testing, creation of fixed distance mount, analysis of mount using cantilever beam approximation to determine minimum required cross-sectional area, integration of potentiometer into brace, development of electronic housing
    - Software – Programming of reproducible procedure capable of outputting patella mediolateral shift to knee flexion angle, reading and analysis of data from Arduino
- Discussed next steps in advancing project
    - Hardware – Calculations of material thickness required
    - Software – Outline potential procedures independently to allow for comparison of ideas

Date: 12.02.26
In Attendance: John Fairhurst, Khaled Gadalla, Emma Graham, Daniel Kekeh, Joshua Lipshaw, Zixuan Zhang
Minutes:

- Discussed brace provided by Professor Stewart for use in the project
- Considered individual ideas for the further advancement of the project

Date: 20.02.26
In Attendance: John Fairhurst, Khaled Gadalla, Emma Graham, Daniel Kekeh, Zixuan Zhang
Minutes:

- Started constructing Arduino system
- Discussed dividing report writing work
- Prioritised tasks for the next week

Date: 24.02.26
In Attendance: John Fairhurst, Khaled Gadalla, Emma Graham, Daniel Kekeh, Zixuan Zhang

Minutes:

- Divided and began testing components
- Tested code on Arduino
- Started report writeup

Date: 02.03.26
In Attendance: John Fairhurst, Khaled Gadalla, Emma Graham, Daniel Kekeh, Joshua Lipshaw, Zixuan Zhang
Minutes:

- Completed section writeups
- Rehearsed presentation
- Tested code on Arduino

Date: 13.03.26

In Attendance: John Fairhurst, Khaled Gadalla, Emma Graham, Daniel Kekeh, Joshua Lipshaw

- Discussed assignment 3
- Laid out steps for progression

Date: 17.03.26

In Attendance: John Fairhurst, Khaled Gadalla, Emma Graham, Daniel Kekeh

- Took hinge apart
- Traced hinge cap
- Finalised CAD design

Date: 22.04.26 – 24.04.26

In Attendance: Khaled Gadalla, Daniel Kekeh

- Iterated prototypes

Date: 28.04.26

In Attendance: John Fairhurst, Khaled Gadalla, Emma Graham, Daniel Kekeh, Joshua Lipshaw, Zixuan Zhang

- Divided work up

Date: 01.05.26

In Attendance: John Fairhurst, Khaled Gadalla, Emma Graham, Daniel Kekeh, Joshua Lipshaw, Zixuan Zhang

- Finalised brochure design
- Discussed improvements to presentation

Date: 04.05.26

In Attendance: John Fairhurst, Khaled Gadalla, Emma Graham, Daniel Kekeh, Joshua Lipshaw

- Practiced presentation